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An Investigation on How to Analyze Design, Implement, Test and Evaluate a GDPR Compliant, Role-Based Secure Web Application

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Authorship Declaration

I, Daniel Ferguson, confirm that this dissertation and the work presented in it are my own achievement.

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Where I have quoted from the work of others the source is always given. With the exception of such quotations this dissertation is entirely my own work;

I have acknowledged all main sources of help;

If my research follows on from previous work or is part of a larger collaborative research project I have made clear exactly what was done by others and what I have contributed myself;

I have read and understand the penalties associated with Academic Misconduct.

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Abstract

The increasing reliance on web applications to store and process personal data has made security and regulatory compliance critical concerns for developers. In particular, the General Data Protection Regulation (GDPR) requires organizations to ensure that personal data is handled securely, transparently, and with appropriate access controls. This dissertation investigates the analysis, design, implementation, testing, and evaluation of a GDPR-compliant web application that incorporates role-based access control (RBAC) to enhance data security and enforce permission-based system usage.

This project involved designing and developing a secure web application using the Python Flask web framework, integrating authentication, role management, and permission assignment mechanisms. The system was designed to restrict access to sensitive functionality based on predefined user roles and permissions, ensuring that only authorized users can perform specific actions within the application. Additionally, GDPR principles such as data minimization, secure storage, and controlled access to personal data were considered throughout the system design.

… More to be written based on testing and evaluation of the site…

This investigation highlights the importance of incorporating privacy-by-design principles and structured access control mechanisms when developing secure web applications that process personal data.

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1 - Introduction

The rapid growth of digital services has resulted in an increasing number of web applications being used to store, manage, and process personal data. From social platforms to enterprise systems, web applications frequently collect sensitive information such as names, login credentials, and personal identifiers. As a result, the protection of user data has become a critical concern for both developers and organizations. Security vulnerabilities in web applications can lead to unauthorized access, data breaches, and misuse of personal information, which may have significant legal and financial consequences.

In response to these concerns, data protection regulations were introduced to ensure that organizations handle personal data responsibly. One of the most significant regulations in this area is the General Data Protection Regulation (GDPR), introduced by the European Union in 2018. GDPR establishes strict requirements for how personal data should be collected, stored, processed, and protected. Organizations that fail to comply with these requirements may face substantial penalties and reputational damage. Consequently, modern web applications must be designed with security and data protection principles integrated from the earliest stages of development (GDPR, 2018).

One important approach to improving application security is the use of role-based access control (RBAC). RBAC is a widely used security model that restricts system access based on user roles rather than individual permissions. By assigning permissions to roles and then assigning those roles to the application’s users, RBAC simplifies access management and helps prevent unauthorized access to sensitive data or system functionality. This approach is particularly relevant in applications where different users require varying levels of access, such as administrators, moderators, and standard users (Lindemulder and Kosinski, 2024).

Despite the availability of security frameworks and best practices, implementing secure and GDPR-compliant web applications remains a challenge for many developers. Many systems fail to correctly restrict user access or properly manage personal data, leaving them vulnerable to misuse or non-compliance with data protection regulations. Therefore, it is important to investigate how secure design principles, structured access control mechanisms, and privacy-focused development approaches can be effectively implemented in modern web applications.

The aim of this project is to design, implement, test, and evaluate a GDPR-compliant web application that incorporates role-based access control to protect user data and system functionality. The project focuses on developing a prototype system that demonstrates how structured permission management and security-focused design can help ensure regulatory compliance and improve application security.

To achieve this aim, several objectives were defined:

- **O1:** To analyse relevant web application security principles, GDPR requirements, and best practices in both web and database design in order to establish a strong theoretical foundation for the system.
- **O2:** To design and implement a web application using the Python Flask framework that incorporates secure authentication, role-based access control, and structured permission management.
- **O3:** To implement additional security features, including two-factor authentication, password management, and protection against common vulnerabilities such as SQL injection.
- **O4:** To develop and integrate supporting system functionality, including user collections and administrative controls, while maintaining secure handling of user data.
- **O5:** To test the system to ensure that authentication, access control mechanisms, and security features operate correctly and prevent unauthorized actions.
- **O6:** To evaluate the completed prototype in terms of its effectiveness, security, usability, and alignment with GDPR principles and the overall project objectives.

The scope of this dissertation focuses on the development of a prototype web application that demonstrates secure role-based access management and GDPR-aware data handling practices. While the system illustrates important security concepts, the world of data security is a fast and ever-changing one, so many of the techniques used will also change and new threats to data security will also change with them. Instead, the project emphasizes demonstrating practical approaches to implementing secure development practices within a manageable academic project.

This dissertation is structured as follows. The following chapter covers the relevant background research and literature related to web application security, GDPR requirements, role-based access control and other data security principles and techniques that any web application would benefit from. From there the chapters will discuss the various prototypes made during the implementation of the project. In each of these chapters, it will describe the methodology and design approaches taken to

make them, present the implementation of the prototype and will explain the technologies and architectures used. They will also outline the forms of testing done to ensure the implemented security mechanisms work as designed. Following that, the evaluation of the final prototype will be discussed, on whether the application meets the effectiveness of its mechanisms and how it compares to the project's original plan. Finally, the last chapter summarizes the findings of the investigation and discusses any future updates and potential improvements on the project and the prototype made.

2 - BackGround - Literature Review

Since the early 2000's websites have been on the rise and are becoming progressively more and more popular daily applications to people of all ages. Modern websites are now being used on a daily basis for all kinds of practices: business, entertainment, communication and even for networking and recruiting. They're used widely in the place of work, allowing business and their employees to have a more comprehensive and fulfilled workplace. According to the research (International Telecommunication Union, 2024), 5.5 billion people are actively using the internet and website. This represents 68% of the global population with an increase of 3.4% every year.

With this increase in popularity comes the increase of user data being stored, used, transferred and read. This has introduced major security concerns, as sensitive information may become accessible to individuals or systems that should not have permission to view or manipulate it. Poorly designed web applications and database systems may expose data to risks such as unauthorized access, data breaches, and cyberattacks. Therefore, it is essential that modern web applications are designed with strong cybersecurity principles in mind, including confidentiality, integrity, and availability.

To combat this in May 2018, the EU passed the GDPR (General Data Protection Regulation) laws. These replaced the previous and now severely outdated Data Protection Directives of 1995. These new laws took into account the new challenges faced by this digital age and focussed on user consent, transparency and put stricter accountability for data processors. This would put definitions of the terms of 'personal data' and outlines rules that are to be followed when processing user data and when and where these rules apply. Another purpose it had was introducing newer principles such as data minimisation, purpose limitation, and secure data processing requirements (GDPR, 2018).

These developments raise an important question regarding the secure development of modern web applications. How can a database-driven web application be designed, implemented, tested, and evaluated in a way that ensures strong security and also being compliant with GDPR regulations? Furthermore, what cybersecurity techniques and design strategies should be applied to protect personal data and prevent unauthorised access? Finally, how can role-based access control be implemented to ensure that different categories of users can only access the information that is relevant and permitted for their role?

This literature review aims to investigate existing research and best practices relating to secure database-driven web development, with a focus on GDPR compliance, role-based security, and the evaluation of security through testing methodologies.

GDPR Compliance

Before discussing how to create this application, it's important that the specific laws we're compliant to should be outlined and analyzed. This is massively important in two regards, these laws apply to all organisations within the EU but also to that of organisations outside the EU that process any personal data of EU citizens. So this means that these laws need to be applied regardless of wherever the development team is or wherever the final system is being hosted (GDPR, 2018). The regulation defines personal data as any information relating to an identified or identifiable individual. This includes names, email addresses, identification numbers, location data, and online identifiers. This definition is significant for database-driven systems, as even seemingly basic account information may fall under GDPR protection.

A key requirement of GDPR compliance is following the data processing principles outlined in Article 5. These principles provide the foundation for lawful and secure handling of personal data and should influence the design of any database-driven web application. The principles introduced here are lawfulness, fairness and transparency, purpose limitation, data minimisation, accuracy, storage limitation, integrity and confidentiality and finally accountability.

Firstly, personal data must be processed lawfully, fairly, and transparently. This means users must be clearly informed about what data is collected, how it is used, and why it is required. Data must also follow purpose limitation, ensuring information is only collected for specific and legitimate purposes and is not reused in inappropriate ways. Additionally, GDPR enforces data minimisation, requiring organisations to collect only the minimum amount of data necessary to provide the service.

GDPR also states that data must be kept accurate and up to date, meaning systems should allow users to correct inaccurate information. The principle of storage limitation requires that personal data is not stored for longer than necessary, meaning retention and deletion policies must be considered in database design. Furthermore, the principle of integrity and confidentiality requires that data is protected against unauthorised access, loss, or damage through appropriate security controls such as encryption and access restrictions.

Finally, GDPR introduces the principle of accountability, requiring organisations not only to comply with GDPR, but also to be able to demonstrate that compliance through documentation, monitoring, and secure system practices.

In addition to GDPR principles, the regulation provides individuals with several legal rights regarding their personal data. These rights significantly influence the required features of a secure database-driven web application.

One of the most important rights is the right of access, which allows users to request a copy of the data held about them. This means web applications must be capable of retrieving and presenting stored data in a readable format.Another key right is the right to rectification, requiring systems to allow users to correct inaccurate information. The right to erasure, may require developers to implement account deletion functionality and ensure data is removed from the database securely. Users also have the right to restrict processing, which means limiting the use of their data without necessarily deleting it. Additionally, the right to data portability requires organisations to provide user data in a structured, commonly used, and machine-readable format. This may influence how data is stored and exported from the database, and may require the development of export tools. These rights highlight that GDPR compliance is not limited to backend security but must also be considered in system usability and user-facing functionality.

One of the main aspects of GDPR is the highlighting of security measures to prevent unlawful access to user data. Article 25 of these laws introduces data protection by design and by default. This means privacy and security must be embedded into the application during development rather than added after completion. Furthermore, article 32 outlines the requirement for appropriate technical and organisational measures to ensure the security of processing. This includes protections such as encryption, access control, confidentiality measures, and ensuring system resilience. In the context of a database-driven web application, this could involve secure password hashing, encryption of sensitive stored data, secure backup procedures, and strict role-based access control mechanisms. Article 32 also encourages regular testing and evaluation of security measures, meaning developers must apply penetration testing, vulnerability analysis, and continuous monitoring as part of the system evaluation process.

Role Based Access

Role-Based Access Control is a widely used cybersecurity technique that restricts system or data access based on the role that is assigned to each user. Rather than permissions being assigned to the users, a predetermined role is created that is assigned the permissions and then assigned further to the user. This provides an extra level of security to a potential web application as users require different levels of access to require different features and/or sensitive data. This is an approach that enforces the least amount of privilege and simplifies the administration in larger systems. (Sonoda, 2020). However, while the majority agree on the core strengths of Role-Based Access Control (RBAC), its effectiveness, how it should be implemented and its main limitations are up for wider discussion.

According to (Park, Sandu and Ahn, 2001) RBAC can be used as a solution to the challenges of access control enforcement on the web. They emphasise that it's valuable in web applications due to its scalability and the wide need for consistent policy enforcement among many users. They highlight that it's not simply a theoretical security model but rather one that can be practically implemented in already established sites through cookies, certificates and directory services. This implementation driven approach contrasts later research which focuses more on how RBAC can be embedded into applications earlier within the software engineering process, rather than that of the deployment stage. This idea is more of a mandatory requirement these days with the inclusion of Article 25 of the GDPR guidelines where it tells us data protection should be designed and considered in early stages (GDPR, 2018).

In comparison, (Mairon Belchior, Schwabe and Parreiras, 2012) argue that one of the main weaknesses that RBAC has is that in practice, access control is often treated as an add-on rather than a core design element. They present RBAC from a model-driven engineering perspective, this suggests that it should be integrated directly into the modelling stage of web application development. Unlike (Park, Sandu, and Ahn, 2001), who prioritise the technical feasibility in the web infrastructure. (Mairon Belchior, Schwabe and Parreiras, 2012), emphasise that the long-term maintainability of RBAC is dependent on if you embed the roles and permissions into the system architecture from the beginning.

Similarly, in (Osborn, 2002) the discussion is expanded further by treating RBAC as a formalised access control framework. This highlights the structural elements such as role hierarchies and separation of duties. While both (Park, Sandu and Ahn, 200) and (Sonada, 2020) focus more on access control as a practical mechanism, Osborn places greater emphasis on it as a model that supports policy consistency and structured organisation security. This suggests that the effectiveness of RBAC is not just based on

restricting user access but also in supporting the organisational governance by preventing conflicts in access rights and privilege misuse.

More recent research shifts its attention towards RBAC relevance in appeasing modern threats and particular insider risks. According to (Marquis, 2024) the effectiveness of RBAC is in reducing insider misuse by limiting user permissions to their role specific requirements. This supports the general claim made earlier that it promotes least privilege, but the article goes further by framing RBAC as a tool for preventing internal data breaches, not just external attacks. In contrast, (Osborn, 2002) emphasis on role hierarchy, the (Marquis, 2024) article highlights that RBAC alone may be insufficient unless supported by monitoring and adaptive security methods. This introduces the idea that RBAC is necessary but not always sufficient within modern day systems.

(Sonoda, 2020) also reinforces the benefits of RBAC in simplicity and scalability in the point of view of a developer. It frames it as a practical approach to reduce management complexity and improve the maintainability in web applications. However, when compared with more research focus sources, the discussion is less concerned with the deeper structural limitations of RBAC such as role engineering complexity or explosion. These are highlighted in more discussion in (Osborn, 2002) and reinforced indirectly by (Mairon Belchior, Schwabe and Parreiras, 2012) through their focus on structured modelling.

Across all sources, a key theme is shared that RBAC does improve security by enforcing least privilege but there are disagreements on what the main challenge it then faces is. (Park, Sandu and Ahn, 2001) imply that it’s implementing it effectively with web tech, whereas (Mairon Belchior, Schwabe and Parreiras, 2012) argue the bigger of the challenges is ensuring that RBAC is designed correctly at the modelling stage. (Osborn, 2002) suggests that the higher difficulty comes in ensuring roles are structured correctly through the hierarchy and the separating of duty constraints. While (Marquis, 2024) states the modern security threats may require that RBAC systems need to be enhanced beyond just that of roles assignment.

Overall, the literature strongly supports Role Based Access Control as an essential security mechanism for a database-driven web application. This is due to its ability to simplify that of the permission management and prevent further unauthorised access. However, the sources collectively suggest that the effectiveness of it depends on not only adopting the model, but also correctly engineering the roles, embedding it early into the system design and supporting it with additional security controls such as monitoring, auditing and secure authentication. These conclusions are particularly relevant when implementing a GDPR compliant system. This is because RBAC contributes to confidentiality, integrity and accountability requirements by limiting data exposure based

on the integrated user roles (Park, Sandu and Ahn, 2001), (Mairon Belchior, Schwabe and Parreiras, 2012).

Authentication and Password Security

Authentication is a massively fundamental security control in web applications. This is due to it determining whether a user is legitimately entitled to access system resources. There is a general consensus that most traditional username and password authentication, on its own, lacks everything needed and presents several vulnerabilities and therefore must be strengthened through more modern security practices.

One main area agreed upon is the necessity for secure password handling and storage. Best practice frameworks emphasise that passwords should never be stored in plain text and must then be protected through strong cryptographic functions combined with unique salts. According to (Gupta, 2025) their framework asserts that adaptive hashing algorithms such as Argon2 or Bcrypt, when paired with unique salts, improve massively against brute force and precomputed hash attacks and advocates for development on policy features such as breach detection to further protect user credentials. This aligns with (OSWAP, 2021) which recommends hashing with salts and strong cryptographic hashing functions, as well as protective measures like login throttling and account lockouts to limit online guessing attacks.

Despite the common agreement these two sources have, there is much divergence in the adequacy and role of multi-factor authentication (MFA). Both (OSWAP, 2021) and (Gupta, 2025) recommend MFA as a substantial enhancement to authentication security, particularly for high risk operations. (OSWAP, 2021) describes MFA as an effective way to mitigate credential compromise by requiring additional proof of identity beyond that of just passwords. While (Gupta, 2025) thinks MFA as a core component of modern authentication strategies is designed to balance usability with stronger assurance. In contrast, (Abdulrahman Al-Mutairi and Rayan Al-Sahli, 2024) highlights the implementation challenges and evolving threat landscape that complicate real world adoption of MFA. For example, recent surveys of current MFA approaches (Abdulrahman Al-Mutairi and Rayan Al-Sahli, 2024) describe how usability issues, insecure recovery mechanisms and environmental complexities can reduce MFA's effectiveness, especially if fallbacks are not securely designed. This suggests that MFA can drastically reduce risk, it introduces new challenges that must be carefully managed.

Another area of discussion appears when comparing the scope and the maturity of authentication approaches. Systematic examinations of multi-factor approaches discuss how biometrics, behavioural and contextual authentication methods can be integrated to further improve security, this looks at future authentication that extends past that of just password plus tokens (Abdulrahman Al-Mutairi and Rayan Al-Sahli, 2024). This forward looking perspective contrasts with more practitioner recommendations. These tend to

focus more on immediate, actionable controls such as password policy enforcement and MFA implementation rather than long term evolution of the authentication architecture.

This source also highlights that authentication is not just a technical mechanism but a usability challenge. While stronger controls like MFA and complex password requirements do improve security, they may also introduce friction that negatively effects the user experience. Both (OSWAP, 2021) and (Abdulrahman Al-Mutairi and Rayan Al-Sahli, 2024) note that usability must be considered alongside security. This is especially because poorly designed recovery or fallback mechanisms can themselves become attack vectors if not securely implemented.

Both again agree on several key principles needed for MFA. Secure password storage is necessary but not sufficient on its own, multi-factor authentication significantly improves resistance against common attacks and effective authentication design must consider the usability as well as technical robustness. However, there is an emphasis between practitioner guidance (OSWAP, 2021) and academic research (Abdulrahman Al-Mutairi and Rayan Al-Sahli, 2024). Practitioner sources prioritise immediate best practices for password handling and MFA implementation, while academic sources provide deeper exploration into the challenges and future directions of authentication systems.

In the context of GDPR compliant web application development, this analysis suggests that authentication systems should not only implement strong hashing and MFA but also be designed with careful attention to usability, recovery processes, and adaptive security enhancements. Addressing these aspects contributes to the integrity and confidentiality of personal data by ensuring that authentication mechanisms resist unauthorized access and support secure user experiences.

Encryption and Data Protection

Encryption is widely regarded as one of the most important technical safeguards when it comes to protecting personal data, particularly in systems where sensitive information may be stored or transmitted across networks. Within GDPR related security, encryption is ever presented as a key measure for reducing the risk of unauthorised access and limiting the impact of potential data breaches. According to (The Cyphere, 2022) this view is supported by emphasising that encryption helps ensure even if an attacker gains access to stored or transmitted data, that information remains unintelligible with the appropriate decryption key. This therefore reflects the GDPR's general expectation that organisations should apply appropriate technical measures to protect personal data depending on risk levels.

A main area of agreement across literature is that encryption must be applied to both data in transit and data at rest. (The Cyphere, 2022) highlights that encryption should be considered across multiple storage and transmission scenarios, including databases, backups, and communication channels, reinforcing that encryption is not only relevant during storage but also during the transferral between the client and server systems. This aligns with broader data protection research which often frames encryption as a foundational security method for preventing unauthorised access in modern digital systems. In contrast, the academic sources tend to focus less on GDPR compliance motivations and more on encryption as a general mechanism for confidentiality and secure data handling. (Das and Samdaria, 2014) highlights encryption as a core component of information security models. This emphasises that confidentiality is maintained when encrypted data cannot be interpreted without access to the correct key material.

However, since GDPR focused sources stress encryption as a compliance and risk mitigation tool, the more technical sources focus on the practical selection of encryption algorithms. (GeeksforGeeks, 2021) provides the idea of Advanced Encryption Standard (AES), they describe it as one of the more widely used symmetric encryption algorithms due to its efficiency, security strength and adaptability through the different keys sizes. They explain that AES is considered highly secure and suitable for modern applications requiring strong encryption protections. Compared to the GDPR guided sources, (GeeksforGeeks, 2021) is less focused on regulatory motivations and more on explaining the internal structure and performance characteristics of AES. This contrast highlights an important difference, GDPR sources primarily justify encryption through legal risk reduction while technical sources justify it through algorithmic strength and implementation feasibility.

Another contrast is how encryption is discussed as part of wider data protection strategies. (The Cyphere, 2022) presents encryption as one component of broader GDPR compliance, this implies that encryption should complement other safeguards such as access controls and organisational security procedures rather than functioning as a standalone solution. This is supported by (Chhetri et al., 2022) which discusses encryption as a safeguard for sensitive personal information and emphasises that encryption must be integrated with wider security approaches such as authentication mechanism and secure system architecture. This reinforces the idea that encryption alone cannot fully protect data unless it's supported by additional layers of defence.

Furthermore, (Chhetri et al., 2022) introduces a further emphasis on the importance of protecting sensitive data due to the severe consequences of breaches. While the sources focused on GDPR highlight encryption mainly in terms of regulatory compliance, (Chhetri et al., 2022) emphasises encryption as essential for protecting highly sensitive records and ensuring privacy and confidentiality in systems where data misuse could lead to significant harm. This reinforces the GDPR focus on protecting individuals' rights and freedoms, but again sees encryption as a safeguard against real world harm rather than that of a legal consequence.

Overall, the sources demonstrate again that encryption is an essential part of protecting personal data. While, the different types of literature have differing reasons to see encryption as an essential force against data breaches with sources, like (The Cyphere, 2022) prioritise the use of encryption to meet that of legal expectations and others like (Chhetri et al., 2022) and (GeeksforGeeks, 2021) seeing it as purely a necessity in protecting high risk personal data. However, like the previous solutions, all sources agree it is not a standalone solution but should be added back up and reinforced with other layers of defense such as the authentication controls, role-based access and a secure system design to provide an effective real world protection.

Incident Response and Logging

Incident response and logging are yet another critical component of a secure web application design, this is particularly when handling personal data. Effective incident response ensures that security events are detected, analysed, contained and remediated in a timely and organised manner. While logging provides the forensic evidence needed to understand the scope of an incident and supports both internal evaluation and regular reporting.

Across a lot of literature there is a general agreement that a formalised incident response programme is essential not only for operational resilience but also for meeting the GDPR related requirements of a web application, such as notifying supervisory authorities of breaches within 72 hours of detection and then documenting the response actions. (Kincaid, 2018) highlights that although GDPR doesn't explicitly require an incident response programme, an organisation cannot comply within the 72 hour notification window without robust detection and response capabilities. This includes clearly defined procedures and trained personnel.

Logging and audit trails are constantly emphasized in both practitioner and research sources as foundational to incident detection and investigation. According to (Datadog, 2022)'s overview of audit logging, it explains the logs record actions taken within a system, captures who, when and what has been done. This is essential for forensic reconstruction and compliance reporting. These logs differ from general application or system logs by focusing on internal actions and state changes that directly inform security analysis and accountability requirements. On the other hand, according to the research (Sharif, 2023), Security Information and Event Management (SIEM) are modern platforms that ingest, normalise, correlate and analyse logs from multiple sources. This provides a unified and real time view of security events that improves threat detection and response effectiveness across complex infrastructures (Granadillo, Zarzosa and Diaz, 2021).

Despite the shared recognition of logging's importance, there is a difference in discussion on how the logs should be used in response workflows. (Sharif, 2023) incident responder guidance highlights detection, containment and recovery activities. They note that effective response depends on continuous monitoring systems and tools that can correlate with log sources in real time. This approach aligns with SIEM on the benefits of mass log analysis for situational awareness, but places more weight on the human and organisational aspects of incident handling, such as communications and coordination during a response.

In addition to detection and analysis the sources distinguish between logging strategies for pre-incident preparedness and post-incident investigation. (ICO, 2024)'s guidance emphasises that organisations must not only detect and manage breaches but also record them systematically, including near misses and details that support trend analysis and future improvements. It recommends maintaining breach logs, reviewing them regularly and implementing retention schedules that balance regulatory needs with data minimisation principles.

In comparison, these sources reveal overlaps and important nuances. Practitioner sided sources like (Kincaid, 2018) and (Sharif, 2023) place strong emphasis on structure response processes, communication plans, organisational readiness and finally, viewing logs that are primarily used to support operational response and reporting. In contrast, more SIEM focused research like (Granadillo, Zarzosa and Diaz, 2021) provide a more technical perspective on logging by detailing how automated correlation, event aggregation and alerting mechanisms can enhance the speed and precision of incident detection, particularly as environments scale in complexity.

A further point of comparison comes in how logging is positioned in comparison to evidence preservation and forensics. While, again practitioner guides stress documentation for regulatory notification and internal learning, research on SIEM and event correlation underscores how logs can be used to reconstruct incident timelines and support analysis. This is critical for demonstrating compliance with GDPR’s accountability obligations, as well as for refining future incident response practices

Overall, the sources both agree on the view that incident response and logging are not optional add-ons but core elements of a secure, GDPR compliant web application. A robust incident response programme requires clearly defined procedures, trained responders and integrated security tooling. While, on the other hand, logging provides the forensic evidence that supports detection, investigation, remediation and regulatory reporting. However, the sources also highlight that technical mechanisms such as SIEM and automated correlation must be paired with organisational readiness, governance and consistent documentation to ensure that incident response is both effective and compliant.

Website Design Considerations

Website design plays a crucial role in creating both effective security and a positive user experience, particularly when personal data protection and regulatory compliance are key requirements. While backend security controls such as encryption, access control and logging are very essential for protecting data, the design of the front-end interface and interaction deeply influences how the users perceive and engage with privacy and security features. The sources suggest that user experience (UX) and privacy and design are not two different concerns. Rather, they are deeply related and must be considered together in a GDPR compliant web application.

A commonality between sources is that privacy and usability must be integrated into the early stages of design, creating a privacy aware user experience. According to (Smashing Magazine, 2019) a framework emphasises that privacy considerations such as data minimisation, transparency and informed consent should be woven into the interaction design, rather than tacked on as an afterthought. This framework focuses on reducing user load and facilitating clear understanding of data collection practices, which in turn supports the GDPR principles of transparency and informed consent. This need for clarity aligns with (Topa and Karyda, 2018) in usability, which finds that security and privacy tools are only effective when users understand them and can use them intuitively. In the absence of usable design, even strong security controls may be bypassed or misused by users who find them confusing or intrusive.

The relationship between usability and privacy is further explored in sources of user perceptions of security tools. The work by (Topa and Karyda, 2018) highlights that user frustration with poorly designed privacy controls, such as unclear warnings or overly complex consent forms, can reduce trust and lead to workarounds that undermine the implement security. This suggests that designers need to prioritise clear visual and interaction cues, contextual help and minimised friction in critical flows such as registration, authentication and consent granting. Such observations are consistent in broader web development guidance. It is recommended that privacy features should be both visible and understandable, this allows users to make educated decisions about their data without feeling overwhelmed.

In more specific design contexts, cookie banners are addressed as a best practice for communicating privacy intent to users while complying with GDPR requirements. (Team, 2022)’s examination of cookie banner UI/UX highlights that consent interfaces should be accessible, but the design should avoid consent fatigue by overwhelming the users with technical details at the outset. This means using clear language, appropriate defaults and easily navigable options that allow users to customize consent without undue effort. This focus on clarity and usability complements the earlier observations

about a privacy aware design and reinforces the idea that front-end elements must support regulatory obligations while sacrificing the user experience.

Another important aspect in website design for privacy is broader data privacy awareness. (Loureiro, 2023) emphasises that user interfaces should help users understand not just what data is collected, but why it is collected and how it will be used. This reflects a shift in design thinking, privacy is not merely a checkbox for legal compliance but an opportunity to build that trust and demonstrate respect for user autonomy. Effective UI design can reduce friction by integrating explanatory elements directly into forms, dashboards and account settings interfaces. For example, contextual tooltips that explain why specific data is needed.

In addition to usability and consent frameworks, practical web development sources emphasise that the front-end design must also support broader security practices. According to (Nucamp, 2024) notes that design choices like form validations, error messaging and session handling interfaces can also influence user trust and perceived security. For instance, error messages that reveal too much internal system detail can disclose security relevant information, whereas clear and consistent messaging can uphold both usability and security goals. This idea resonates with the theme that privacy and security must be considered entirely in user interaction design rather than as isolated features.

Across the sources, two themes are constant. First, clarity and intuitiveness in design are fundamental for translating complex privacy and security requirements into user actions that are both understandable and manageable. Whether dealing with cookie consent banners, form interactions or privacy notices, users need interfaces that communicate not only what is happening but why it matters. Second, website design must balance regulatory compliance and user empowerment, this means making privacy protections visible, optional and controllable without overwhelming or confusing the users.

In summary, the sources position website design as both a supporting structure for security controls and a driver of user behaviour that then influences how privacy and security features are used in practice. Effective website design for GDPR compliant systems require privacy aware frameworks, clear consent mechanisms, user-centred interaction patterns and thoughtful messaging that together support legal requirements, secure practices and an overall positive user experience outcome.

Database Design and SQL Injection Prevention

Database design and the prevention of SQL injection vulnerabilities are another foundational component of building a secure web application that can handle sensitive user data. SQL injection remains one of the most prevalent and damaging forms of attack. It enables the attackers to manipulate the database queries to access, modify or destroy data beyond authorised permissions. The reviewed sources collectively agree on the importance of secure database architecture, input validation and query handling as central strategies to prevent these attacks.

A constant theme across the sources is that an insecure handling of user input is the primary root cause of SQL injection vulnerabilities. According to (OSWAP, 2025) SQL injection occurs when untrusted input is improperly included in dynamic SQL queries, allowing attackers to inject malicious code that alters logic and then interacts with the database in unintended ways. This source advocates for defensive coding practices such as parameterised queries and prepared statements, which therefore separate the code logic from the user data. Similarly, (Cloudflare, 2025) highlights how malicious input can alter the structure of the database commands and outlines that sanitation and use of secure APIs are key measures in preventing such attacks. Both of these sources align on the importance of treating all user input as potentially hostile and encoding or escaping it appropriately before inclusion in database interactions.

Academic research reinforces these practitioner ideas by emphasising on secure query handling but also points to broader architectural and systemic concerns. The review of SQL injection prevention notes that while parameterised statements and input filtering are widely recommended, prevention must be considered across the full stack, that includes application logic, database schema design and deployment environments. It highlights that effective prevention requires a combination of secure programming practices, automated scanning tools and runtime protections that can detect and block exploit attempts in real time (Hazaline Johny et al., 2021). This multiplatform perspective expands on the more code centric guidance of (OSWAP, 2025) by situating SQL injection prevention within the context of system design, suggesting that developers must consider not only query construction but also system behaviour under attack.

In contrast, a study on database design patterns and security (Paul, Sharma and Olukoya, 2024) emphasises how secure scheme design itself can reduce the likelihood and impact of injection vulnerabilities. This research argues that when database schemas and relationships are carefully normalised and restrictive constraints are applied, chances of an SQL injection is reduced because poorly structured tables and unvalidated fields are less prevalent. It points to practices such as using explicit data

types, enforcing integrity and separating business logic from data access layers as part of a defensive architecture that discourages insecure coding patterns. Compared to (OSWAP, 2025) and (Cloudflare, 2025) which focus mainly on runtime and input handling, this academic work emphasises how database structure influences vulnerability likelihood at a deeper level.

Another discussion to be had is the automation of detection and testing. According to (Nasereddin et al., 2021) the value of automated vulnerability scanners and analysis tools for identifying SQL injection risks early in the development cycle, including static code analysis, fuzz testing and real time monitoring. While these tools are not replacements for secure coding practices, they act as complementary mechanisms to detect overlooked weaknesses and provide feedback for iterative improvements. This aligns with the broader sources that suggest secure application development must incorporate test and evaluation as part of the software development lifecycle, rather than relying purely on coding standards.

Although the sources generally agree on core prevention strategies, there are subtle differences in emphasis. More practitioner focused sources like (OSWAP, 2025) and (Cloudflare, 2025) concentrate on specific coding techniques that developers can apply immediately to reduce SQL injection risk. While, academic sources like (Hazaline Johny et al., 2021) extend the discourse to include system level architecture, testing frameworks and schema design as important contributors to overall database security, For example, the research on SQL injection prevention highlights multi-layered strategies that not only reduce the chances of vulnerability but also offer mechanisms for active detection and response, such as deployable intrusion detection systems and query sanitisation.

Another point of comparison is the extent to which the sources address the impact of SQL injection in a GDPR compliant system. While again, (Cloudflare, 2025) and (OSWAP, 2025) focus on attack mechanics and prevention tactics, the academic sources like (Paul, Sharma and Olukoya, 2024) often situate SQL injection within the broader discussion of data integrity, accountability and privacy implications while noting that unauthorized data access is directly contrary to core GDPR principles of confidentiality and integrity. This is particularly relevant for systems that handle personal data. Breaches enabled through injection are not merely technical faults but represent potential violations of regulatory obligations.

Overall, the sources cover several key principles for database design and SQL injection prevention. Treat all user input as untrusted and apply sanitisation and validation consistently (OSWAP, 2025: Cloudflare, 2025). Use parameterised queries, prepared statements and secure APIs to separate code logic from data (OSWAP, 2025). Consider

architectural and schema design as part of risk reduction, including referential integrity and strict typing (Paul, Sharma and Olukoya, 2024). Integrate automated scanning and testing tools into the development lifecycle to detect flaws early (Nasereddin et al., 2021). Finally, recognise that SQL injection vulnerabilities have both technical and regulatory implications, especially in GDPR contexts where unauthorized data exposure undermines legal compliance (Paul, Sharma and Olukoya, 2024). By combining these perspectives, from immediate code level tactics to more general architectural considerations, developers can build database systems that are resilient against injection attacks and aligned with secure, GDPR compliant web application practices.

Testing and Evaluation

Testing and evaluation are essential components of a secure web application development. They provide evidence that implemented security controls actually function as intended and that systems comply with both technical and regulatory requirements. In the context of a GDPR compliant application, testing processes such as penetration testing, automated security testing and formal evaluation frameworks that help identify vulnerabilities, verify compliance and support accountability obligations of Article 32 (GDPR, 2018). Across the sources, there is broad agreement that comprehensive testing must combine both automated and manual approaches, integrate with development life cycles and extend beyond functional correctness to assess security posture.

According to (Wipro, 2021), a foundational position on testing within secure development frameworks is testing as part of a structured lifecycle that includes requirement analysis, secure design, implementation, verification and validation. The framework emphasises that security testing shall not be isolated to late stage validation but should be integrated at multiple stages to identify defects early and reduce remediation costs. Specifically, techniques such as static and dynamic analysis, code reviews and automated test scripts are recommended to ensure both functional behaviour and non-functional security properties are evaluated. This integrated view aligns with broader engineering practices that advocate continuous testing rather than end of cycle assessment.

Penetration testing is frequently highlighted as a key evaluation method for web application security, especially for detecting vulnerabilities that automated tools may miss. (Rhino Security Labs, 2018)'s guide of GDPR and penetration testing explains how ethical hacking exercises simulate real attacker behaviour to uncover security weaknesses in authentication, access control, input validation and session management. Importantly, penetration testing can help organisations demonstrate compliance by showing that they have actively sought out and mitigated vulnerabilities, which is valuable evidence under GDPR's accountability requirements. This view is supported by (Varghese, 2026)'s web application penetration testing overview, which positions penetration testing as a targeted evaluation that complements automated scans by exploring complex attack paths and business logic vulnerabilities that automated tools cannot easily detect.

While penetration testing focuses on attacker simulation, automated testing, including unit, integration and regression tests, provides a complementary mechanism for ensuring ongoing correctness and security of application behavior. (Geeksforgeeks, 2019) discusses software testing and how automated tests can improve test coverage

and consistency, enabling repeated execution of security test cases during development and deployment cycles. Automated testing frameworks can integrate with continuous integration/continuous deployment (CI/CD) pipelines to catch regressions early and validate that security controls remain effective as code evolves.

Academic research expands on these sources by examining formal testing methodologies and their effectiveness. (Hilario et al., 2024) proposes structured security testing models that extend beyond ad-hoc penetration testing, incorporating formal test case generation, risk based prioritisation and systematic vulnerability classification. It argues that such formalised testing approaches can provide more consistent and reproducible results, as well as measurable security indicators that support continuous improvement of security posture. This academic view highlights that without structured evaluation metrics and classification systems, testing outcomes may remain unreliable rather than driving systematic enhancement of security controls.

A further comparison emerges in research that evaluates the practical impact of testing and secure coding practices on real systems. The (Wen and Katt, 2023) article on secure software engineering provides evidence that rigorous testing, including threat based evaluation, code analysis and exploitation attempts, correlates with reduced vulnerability in production applications. It also discusses the need for security regression tests that can automatically check for reintroduction of previously fixed vulnerabilities, an approach that mirrors the incremental nature of modern agile development practices. This emphasis on regression testing complements both penetration testing and automated unit tests by focusing on change related risk.

There is a common agreement across sources that testing activities should be contextualised within the organisation’s risk profile and regulatory obligations. For GDPR complaint systems, this means not only identifying and mitigating technical vulnerabilities but also verifying that privacy mechanisms, such as consent flows, access controls and data minimisation features, function correctly and do not undermine legal requirements. (Rhino Security Labs, 2018) suggests that GDPR specific penetration testing may include scenarios such as testing unauthorized access to personal data, escalation of privileges and attempts to bypass consent mechanisms, tests that are not always included in generic security assessment frameworks.

Despite the agreement on the importance of testing, the sources vary in emphasis. Practitioner sources like (Wipro, 2021; Geekforgeeks, 2019) focus on integration with development processes and continuous quality assurance, whereas penetration testing guides emphasize attack simulation, human driven exploration and evidentiary value for compliance demonstration (Rhino Security Labs, 2018; Varghese, 2026). Academic work tends to concentrate on methodological rigor, structured metrics and formal test

specifications (Hilario et al., 2024; Wen and Katt, 2023). Together, these views highlight that a mature testing strategy for secure applications should balance multiple dimensions, automated checks for consistency and regression, structured formal methods for measurable security indicators and manual penetration testing for uncovering complex vulnerabilities.

In summary, the sources identify several core principles for effective testing and evaluation in a secure, GDPR compliant web application. Integration with the development lifecycle, security testing should be embedded through the cycle rather than performed only at the end (Wipro, 2021; Geeksforgeeks, 2019). Diverse testing techniques, a combination of automated testing, formal security evaluation and penetration testing is required to address different classes of vulnerabilities (Rhino Security Labs, 2018; Varghese,2026; Hilario et al., 2024). Contextualised evaluation, testing must consider both technical vulnerabilities and regulatory requirements, especially regarding personal data access and privacy features (Rhino Security Labs, 2018). Finally, measurable outcomes, formalised testing approaches that generate reproducible security metrics support continuous improvement and compliance demonstration (Hilario et al., 2024; Wen and Katt, 2023).

Conclusion

This literature review has investigated how a secure, GDPR compliant, role-based web application can be effectively analysed, designed, implemented, tested and evaluated. Across the reviewed literature, there is strong agreement that building such an application requires a structured security approach that integrates both technical controls and regulatory compliance requirements throughout the full software development lifecycle. Rather than treating GDPR compliance as an additional feature applied after development, the research indicates that privacy and security must be embedded into early analysis and design decisions, supporting GDPR principles such as accountability, integrity, confidentiality, transparency and data minimisation.

The sources demonstrate that effective analysis and planning are essential foundations for GDPR compliant development . Establishing the types of personal data being processed, the legal basis for processing, and the system's risk profile informs what technical controls must be prioritised. GDPR compliance is not achieved through one single technology, but through a combination of secure processing practices, documentation and security measures that protect personal data and demonstrate responsible data governance.

Role-Based Access Control (RBAC) is consistently highlighted as a key mechanism for ensuring that users only access the minimum information required for their role. The review sources indicate that RBAC improves confidentiality and reduces insider risk by limiting privilege levels and enforcing structured permission hierarchies. However, the literature also suggests that RBAC must be implemented carefully to remain scalable, testable and maintainable as web applications evolve. This reinforces the importance of integrating RBAC into both design and database architecture rather than implementing it solely through application logic.

Authentication and password security are also shown to be essential to protecting user accounts and preventing unauthorized access to personal data. The sources strongly support the use of secure password storage mechanisms such as modern hashing algorithms and the adoption of multi-factor authentication to mitigate credential theft and password based attacks. However, the research also identifies that security mechanisms must be usable and well designed, as poorly implemented authentication flows and insecure recovery mechanisms can undermine even strong security models.

Encryption and data protection strategies are widely recognised as necessary safeguards for protecting sensitive data both at rest and in transit. The literature supports encryption as a major control that reduces the impact of data breaches, while also indicating that encryption alone is insufficient unless it is supported by strong

access control, secure key management, and secure system architecture. This reinforces the idea that secure implementation requires layered defence, where multiple security controls work together to prevent and mitigate compromise.

From a design perspective, the research indicates that GDPR compliance is strongly influenced by the usability and transparency of the website itself. Privacy aware user experience, clear consent mechanisms and accessible privacy controls are presented as important design considerations, as they shape user trust and support informed decision making. This suggests that front-end website design is not separate from security but is a major factor in whether privacy mechanisms are effective in a real world use.

Database design and SQL injection prevention are identified as critical for protecting stored personal data, particularly since injection vulnerabilities can allow attackers to bypass authentication and access control entirely. The reviewed sources strongly support the use of parameterised queries, prepared statements, and secure database architecture, alongside automated scanning and testing to prevent injection vulnerabilities. This highlights that secure database implementation must be approached both through technical query handling controls and through secure structural design.

Finally, the literature strongly supports the view that testing and evaluation are essential to confirming that security measures are correctly implemented and remain effective over time. Automated testing, vulnerability scanning and penetration testing are repeatedly presented as key methods for evaluating security posture, while academic research highlights the need for structured evaluation frameworks and measurable testing outcomes. Within GDPR driven development, testing also serves a compliance function, as it provides evidence that appropriate safeguards are in place and actively maintained.

Overall, the reviewed sources suggest that the most effective method for developing a GDPR compliant, role-based secure web application is through an integrated approach that combines, structured GDPR analysis, privacy aware design, RBAC enforcement, strong authentication mechanisms, encryption based protection, secure database architecture and continuous testing and evaluation. While each security component contributes individually to system resilience, the literature emphasises that the strongest protection is achieved through in-depth defence, where multiple controls operate together to prevent, detect and respond to threats. Therefore, it can be concluded that GDPR compliance and secure implementation are not achieved through isolated technical solutions, but through a lifecycle driven development process that embeds security privacy and evaluation into every stage of the system’s design and operation.

3 - Methodology

The concept of the application is to provide a platform for collectors of action figures and statues, functioning as a hybrid between a wiki and a personal collection management system. Users are able to search for figures, access detailed information about items they do not own, and maintain a personalized digital record of their own collections. This includes the ability to track individual items as well as estimate the overall value of a collection. While these features are primarily user-focused, they also introduce challenges related to data handling and access control, particularly when personal user data is involved.

To support the structured development of the application, a set of system requirements was defined and documented within a requirements table (see Appendix 1). Each requirement is assigned a unique identifier and accompanied by a detailed description outlining its intended functionality. This approach provides a clear and organised method for defining system behaviour, ensuring that all features are explicitly specified and can be consistently referenced throughout the development process. Furthermore, this structure supports traceability, allowing requirements to be linked to specific stages of analysis, design, implementation, and testing. This ensures that all aspects of the system can be evaluated against their original objectives and reduces the risk of missing or incomplete functionality.

An iterative prototyping approach was adopted for the development of the system. An initial prototype was created prior to extensive background research in order to establish a functional baseline and provide practical insight into the problem domain. This early implementation helped to identify key challenges and informed the direction of subsequent research. Following this, additional prototypes were developed, each building upon the previous version and incorporating findings from the research phase. This approach allowed for continuous refinement of both functionality and security features, ensuring that improvements were guided by both practical experience and theoretical understanding.

This iterative methodology also enabled flexibility in decision-making, particularly when introducing new technologies and security measures. For example, changes such as the transition to a more robust database system and the inclusion of enhanced security features were informed by insights gained throughout the development process. By progressively refining the system across multiple iterations, the project was able to balance functionality, usability, and security, ultimately working towards the development of a GDPR-compliant web application.

In terms of implementation technologies, one primary tool was selected for the initial prototype. The application was developed using the Flask web framework in Python, largely due to prior experience and familiarity. This decision enabled rapid development and access to a wide range of extensions, particularly those related to authentication and security. Flask's lightweight and modular nature also makes it well-suited for prototyping, allowing features to be incrementally added and refined.

4 - Prototype 1

Analysis

The initial prototype of the system was designed and implemented prior to conducting extensive background research. This approach was intentionally adopted to provide practical insight into the problem domain and to identify key areas requiring further investigation. By developing an early working version of the application, it was possible to better understand the challenges associated with implementing secure web applications and to inform subsequent research decisions. This iterative approach aligns with agile development principles, where early prototypes are used to refine requirements and guide future development.

For the first prototype, SQLite was chosen as the database management system for the initial prototype. This decision was based on its simplicity, ease of integration with Flask, and again with prior experience with the technology. SQLite is widely used for small-scale applications and prototyping, and its minimal configuration requirements allowed for efficient early development. However, it is acknowledged that SQLite has limitations in terms of scalability and advanced security features. As a result, future iterations of the system may transition to a more robust database system such as MySQL. Early research showed that MySQL provides more extensive support for security-related features and better aligns with the long-term goals of developing a secure and scalable application (see Figure 1).

	Encryption at Rest	Encryption in Transit	Role-Based Access Control	Audit Logging	Row Level Security	Built In Authentication	Data Masking/ Redaction
MySQL	Yes	Yes	Yes	Yes (with plugins)	No	Yes	No
PostgreSQL	Yes	Yes	Yes	Yes	Yes	Yes	Yes
SQLite	Yes (with extensions)	No	No	No	No	No	No
Microsoft SQL Server	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Oracle Database	Yes	Yes	Yes	Yes	Yes	Yes	Yes
MongoDB	Yes	Yes	Yes	Yes (with paid version)	Yes (with paid version)	Yes	No
MariaDB	Yes	Yes	Yes	Yes	Yes	Yes	No
Firebase Firestore	Yes	Yes	Yes	Yes	No	Yes (with paid version)	No

Figure 1 - Table of database softwares and the security features they include.

The functionality of the initial prototype was guided by a set of core requirements, referred to as “Must Have” features. These features represent the essential components that must be present in all versions of the system, particularly those relating to security and access control (see Figure 2). By establishing these requirements early in the development process, the project ensures that security remains a consistent priority throughout all stages of implementation.

1	Python site with a linked DB	This is really the main requirement, this is the site. This should be a html/Python Flask website with a database linked to it
2	Account signup/login with basic security	This should allow the user to signup an account and login to it. The security with this means it should have requirements to meet in terms of the user info (mainly the password). Also means that attackers can't just log into any account, a password of the users choosing is required.
3	The creation of user roles and permissions	This should allow a user (later down the line with the correct permissions) to create roles and permissions within the website
4	Assign permissions to roles and roles to users	This will allow for already existing perimissions to be assigned to already existing roles within the site. Users can then be assigned roles and should then be able to perform actions with the roles permissions

Figure 2 - Must Have's that should be featured in every prototype.

In addition to these core requirements, the prototype also included several non-security-related features necessary for the overall functionality of the application. These include search and navigation capabilities, as well as features supporting the management of user collections (see Figure 3). While these features are not directly related to security, they are essential for providing a functional and user-friendly system, and they introduce additional considerations for how user data is stored and accessed.

5	Include encryption and then revelent decryptions	User 's sensitive info should be encrypted within the db. This info should still be encrypted when appearing on the site. However, if user's with relevant permissions try to access this info, it should then be decrypted and displayed to them. This should only be relevant info to certain tasks.

9	Navigation Features	This should come as a nav bar and search bar. This should allow the users to easily navigate the site and search for figures and others accounts within the site.
10	Add Figures, Brands, Manufacturers and Companies	this is a form that allows for action figures and their brands and manufacturers to be add to the sites DB. These can then be searched for in the sites search bar.

Figure 3 - Prototype 1's "Should Have" Features

The "Could Have" features include multi-factor authentication (MFA) and incident reporting mechanisms. MFA was identified as a valuable enhancement to improve account security. However, it introduces additional implementation complexity and was therefore deferred to later prototypes. Similarly, incident reporting and response functionality would support security monitoring and GDPR-related breach handling, but requires more advanced infrastructure than was appropriate for the initial prototype (see Figure 4).

The "Won't Have" features for this stage include explicit SQL injection prevention measures and front-end design improvements. While protection against SQL injection is critical, the use of parameterised queries within Flask provides a basic level of security, with more comprehensive mitigation to be implemented in later iterations. Front-end design was also deprioritised, as the focus of the prototype was on core functionality rather than user interface refinement (see Figure 5).

While these features have been determined in depth for the initial prototype they will be implemented in later versions of the site. As aforementioned, the purpose of the first prototype was to get insight of the problem and highlighted areas for later research.

6	Two Factor Authentication	When a user signs up an account they should have to link a email to the account. When the account is created they will be sent an 2fa email with a code to verify their identity. This will occur again when the user tries to login to their account too.
7	Incident Response and Logging	When a user fails to login to their account, a log should be made, indicting when the failed attempt was made. If the user fails 3 times the account should be timed out and an email sent to the user indicating an attempt was made.
8	Change Password	In the emails sent to the user indicating an attempt to login to their account was made, there should be a link that allows the user to change their password for security purposes.

Figure 4 - The “Could Have’s” of prototype 1

11	SQL Injection Prevention	Validating and sanitizing user inputs and not inputting them directly into the DB will help prevent SQL injection attacks.
12	Front-end Website Design	This is simply how the site will look and navigate. It will allow for clear and easy usage of the security features. This clear usage will prevent breaches and allow usability within the site.

Figure 5 - The “Won’t Have’s” of prototype 1

Design

The design of Prototype 1 focused on implementing a foundational role-based web application with basic authentication and access control mechanisms. The system was developed using the Flask framework, following a modular structure that separates routing, data models, and presentation through templates. This approach allows for clear organisation of functionality and supports future extensibility.

At the core of the design is the database structure, which was planned using a relational model implemented through SQLAlchemy and SQLite. The primary entity identified was the User, which would store account credentials alongside personal data. Due to the sensitive nature of this information, it was decided that personal attributes such as name, email, date of birth, and country should be encrypted at rest. Additionally, passwords were designed to be stored using secure hashing rather than plaintext, ensuring that user credentials would be protected even in the event of a data breach. To support role-based access control, the system design included separate Roles and Permissions entities, linked through a many-to-many relationship. This structure allows roles to be assigned multiple permissions, and users to inherit permissions through their assigned role. This design was chosen to provide flexibility and scalability, enabling permissions to be managed independently of individual users (see Figure 6).

A key aspect of the design is the concept of conditional data access. It was proposed that user information displayed within the system should vary depending on the permissions of the current user. Sensitive fields, such as personal details, would only be visible to users with the appropriate permissions, while other users would see restricted or redacted data. This approach ensures that access to personal information is controlled in accordance with security and data protection principles. Furthermore, authentication was designed as a session-based system, where users would log in using a username and password, and their authenticated state would be maintained through session variables.

Overall, the design of Prototype 1 prioritised the establishment of a clear and extensible structure, with a particular focus on role-based access control and the protection of sensitive user data. While the design remains relatively simple, it provides a strong foundation for the integration of more advanced security features and GDPR-compliant mechanisms in further prototypes.

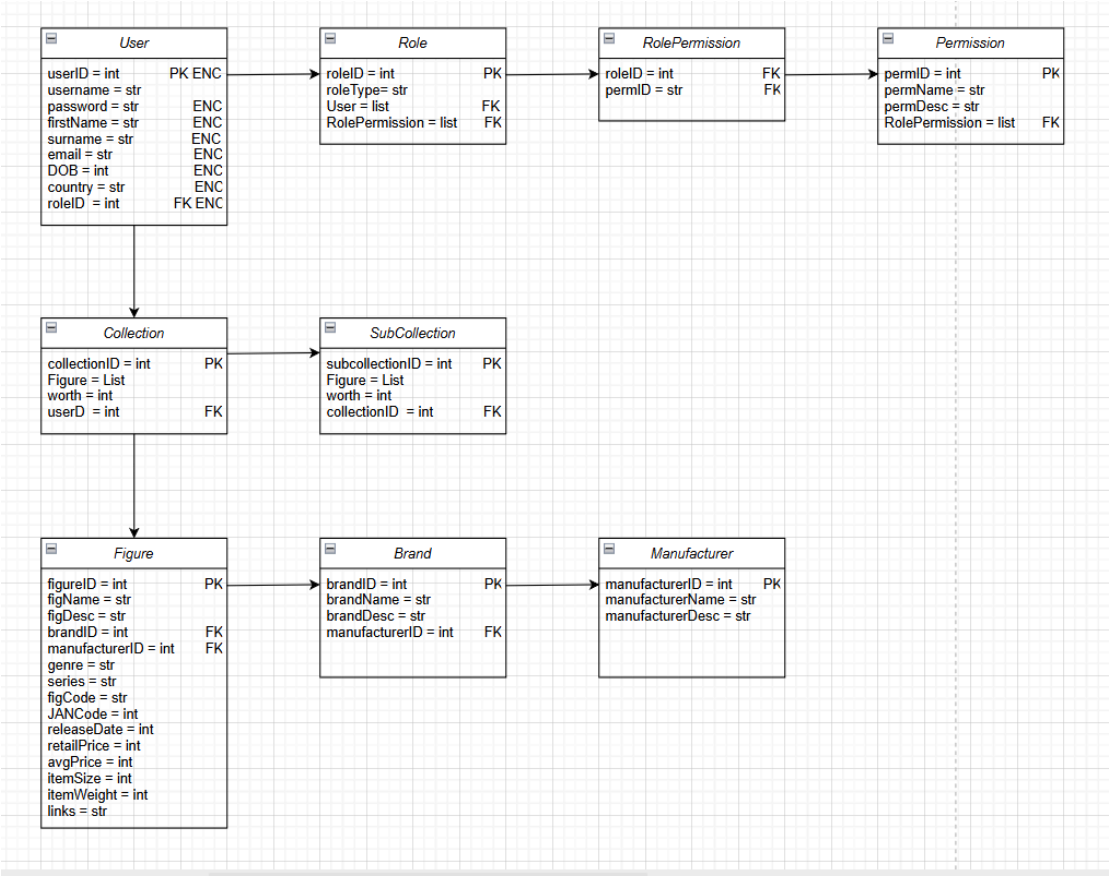

Figure 6 - Database relationship diagram. Shows the primary entity of the “Users” table how it links to everything else in the DB.

Implementation

The implementation of Prototype 1 was carried out using the Flask web framework in Python, with SQLAlchemy used as the Object Relational Mapper (ORM) to manage interactions with an SQLite database. This combination allowed for rapid development while maintaining a clear structure between application logic and data management.

User authentication was implemented using a username and password system. Passwords were securely stored using hashing provided by the Bcrypt library, ensuring that plaintext credentials were never saved within the database. Upon successful login, session variables were used to maintain the user's authenticated state, enabling access to protected areas of the application. In addition to authentication, encryption was implemented for sensitive user data. Fields such as surname, forename, email, date of birth, and country were encrypted before being stored in the database using symmetric encryption. Decryption was performed when the data needed to be accessed within the application. This approach ensured that personally identifiable information was protected at rest, aligning with data protection requirements.

The role-based access control system was implemented through three primary database models: Roles, Permissions, and RolePermissions. These models established a many-to-many relationship between roles and permissions, allowing permissions to be dynamically assigned. Functionality was developed to allow administrators to create roles and permissions, assign permissions to roles, and assign roles to users through dedicated routes and forms within the application. A key feature of the implementation was the enforcement of conditional data access. This was demonstrated through the users page, where the visibility of sensitive user information depended on the permissions of the currently logged-in user. If a user lacked the required permission, fields such as personal details were replaced with redacted values. Conversely, users with the appropriate permissions were able to view the full dataset. This provided a practical demonstration of role-based access control within the application.

Generally, the implementation of Prototype 1 successfully established core functionality, including authentication, encrypted data storage, and role-based access control. While the system remains limited in terms of advanced security features and scalability, it provides a working foundation for further development and evaluation in subsequent prototypes. However, one discovery made was that all permissions and their purpose would need to be made prior to implementation. This is since for the permissions to work their "PermissionID" has to be hardcoded into the feature it's using. So in further prototypes, the permissions will have to be in the form prior so their "ID's" can be coded into the security of the site.

Testing

Testing of Prototype 1 was conducted to ensure that the core functionality of the system operated as expected, particularly in relation to authentication and role-based access control. Given the relatively small scale of the prototype, testing was primarily carried out using manual and functional testing methods.

Authentication functionality was tested by attempting to create user accounts with both valid and invalid input data. This included testing for duplicate usernames, mismatched passwords, and insufficient password length. These tests confirmed that validation checks were correctly enforced, preventing invalid account creation. Login functionality was also tested using both correct and incorrect credentials to ensure that access was only granted to authorised users and that appropriate error messages were displayed when authentication failed (see Appendix 2).

The role-based access control system was tested by creating a role and then respected permissions, assigning these permissions to roles, and then assigning roles to users. The behaviour of the system was then observed when accessing restricted content. In particular, the users page was used to verify conditional data access. Users without the required permissions were only able to view limited information, with sensitive fields redacted, whereas users with the appropriate permissions were able to view full user details. This demonstrated that permission checks were functioning correctly and that access control was being enforced as intended (see Appendix 3).

Basic testing of data protection mechanisms was also performed. This involved verifying that sensitive user data stored in the database was not readable in plaintext and was correctly decrypted when accessed through the application. Password storage was also reviewed to confirm that hashing had been applied (see Figure 7).

surname	forename	email
Filter	Filter	Filter
gAAAAABpqEkGURECv4I8cQ_f7dX1fgAXOm5R…	gAAAAABpqEkGOr6MuTpUDgDfw9C6sNMkXPi6…	gAAAAABpqEkGvggJ2CHwzJZ0Y3Gj9I2GPAzf…

country	bio	password
Filter	Filter	Filter
gAAAAABpqEkGdYAbzoRQJSeGef1x8CxXsi5X…	NULL	$2b$12$Jrr.EAhR6xNPDSuy1M4LqOhz.MZT5…

Figure 7 - Confirmation of User Info Encryption within the Site's Database.

Overall, testing of Prototype 1 confirmed that the core features of the system were functioning correctly. However, testing was limited in scope and primarily manual, meaning that more comprehensive testing methods, such as automated testing and

security-focused testing, will be required in later prototypes to ensure robustness and resilience against potential vulnerabilities.

5 - Prototype 2

Analysis

The development of Prototype 2 was informed by the findings of the background research conducted after the completion of the initial prototype. While Prototype 1 primarily focused on establishing a functional foundation, this iteration aimed to significantly enhance the system with a stronger emphasis on security, scalability, and alignment with GDPR principles. As a result, several key areas identified during research were incorporated into the design. These included improvements to authentication and password security, such as the enforcement of stronger password policies and more secure credential handling. Enhanced encryption and data protection mechanisms to ensure that sensitive user data is securely stored and processed. The introduction of incident response and logging strategies to enable the detection and monitoring of potential security events. In addition, greater consideration was given to website design, particularly in relation to usability, accessibility, and the secure handling of user input within the interface. The database design was also revisited, with a focus on improving structure, scalability, and security through the adoption of a more robust database management system and the implementation of measures to prevent SQL injection attacks. Collectively, these additions demonstrate a shift from a purely functional prototype to a more comprehensive, security-focused system guided by established best practices and regulatory requirements.

To meet those new changes to database requirements, the decision was made to transition from SQLite to MySQL as the database management system for Prototype 2. While SQLite was suitable for the initial prototype due to its simplicity and ease of integration, it was identified through research, back during prototype 1 (see Figure 1), as limited in terms of scalability, concurrent access handling, and advanced security features. As the system evolved to incorporate more complex data structures and security requirements, a more robust solution was necessary. MySQL provides improved support for multi-user environments, stronger access control mechanisms, and better compatibility with security-focused configurations. Additionally, it offers greater flexibility in managing larger datasets and supports more advanced database design practices. This transition reflects a move towards a more scalable and secure architecture, better suited to handling sensitive user data and aligning with the overall aim of developing a GDPR-compliant web application.

The “Must Have” requirements for Prototype 2 remained consistent with those established in Prototype 1 (see Figure 2). These core features, particularly user authentication and role-based access control, continued to form the foundation of the system. Maintaining these requirements ensured continuity while allowing further

development to focus on enhancing security, functionality, and overall system robustness without redefining the fundamental objectives of the application. Furthermore, since tests from prototype 1 show the implemented features work (see Appendix 1), these features can just be added to.

The “Should Have” requirements for Prototype 2 focused on enhancing both the security and usability of the system, building upon the foundational features established in the initial prototype. Key security improvements included the introduction of two-factor authentication to strengthen the login process, as well as the implementation of incident response and logging mechanisms to enable the monitoring and tracking of system activity. A password management feature was also identified as a priority, allowing users to securely change their credentials when required. In addition, measures to prevent SQL injection were incorporated through the use of parameterised queries and input validation, improving the overall security of the application. Alongside these security enhancements, several usability and functionality improvements were included. These consisted of the development of a more refined front-end design to improve user experience, as well as the introduction of collection and sub-collection features, allowing users to better organise and manage their figures. Furthermore, an admin interface was proposed to support system management tasks such as overseeing users, roles, and permissions. Collectively, these features represent a significant step towards a more secure, user-friendly, and fully functional web application (see Figure 8).

6	Two Factor Authentication	When a user signs up an account they should have to link a email to the account. When the account is created they will be sent an 2fa email with a code to verify their identity. This will occur again when the user tries to login to their account too.
7	Incident Response and Logging	When a user fails to login to their account, a log should be made, indicting when the failed attempt was made. If the user fails 3 times the account should be timed out and an email sent to the user indicating an attempt was made.
8	Change Password	In the emails sent to the user indicating an attempt to login to their account was made, there should be a link that allows the user to change their password for security purposes.
9	Navigation Features	This should come as a nav bar and search bar. This should allow the users to easily navigate the site and search for figures and others accounts within the site.
10	Add Figures, Brands, Manufacturers and Companies	this is a form that allows for action figures and their brands and manufacturers to be add to the sites DB. These can then be searched for in the sites search bar.
11	SQL Injection Prevention	Validating and sanitizing user inputs and not inputting them directly into the DB will help prevent SQL injection attacks.

12	Front-end Website Design	This is simply how the site will look and navigate. It will allow for clear and easy usage of the security features. This clear usage will prevent breaches and allow usability within the site.
13	Change of Project Layout	Changing the project to three tier architecture will provide the project with enhanced scalability, security and an overall faster development cycle due to having features and functions being separated.
14	Implement of permissions into the site	Permissions and their respected ID's have to be coded into the site. This is so that the features can check if the user has the correct permissionID before letting them perform it.
15	Collections and Sub-Collections	This is the ability to let users add figures from the site to their own personal collection. This should provide a value of their overall collection. They should also be able to make sub-collections which can be smaller collections based on displays they have.
18	Input data into the DB	This simply filling out the figures, brands and manufacturers tables in the db with existing data.
19	Admin Pages	This should be a page only accessible to those with an admin role that allows them to perform admin tasks. E.g. Suspend/unsuspend accounts, reset account passwords, etc.

Figure 8 - The “Should Have’s” of Prototype 2

The “Could Have” requirements for Prototype 2 focused on additional user-focused features that would enhance personalisation and social interaction within the application. These included extended account management functionality, allowing users to edit and personalise their profiles, as well as the introduction of a friends system, enabling users to connect with others on the platform. While these features would improve user engagement and overall experience, they were not considered essential to the core objectives of the prototype, particularly in relation to security and GDPR compliance, and were therefore given lower priority.

16	Account Features	This is simply the ability for users to edit and customise their accounts based on their own preferences. This should also allow them to permanately delete their account if they so choose to.
17	Friend Features	Users should be able to search for other accounts on the site and add them to their friends list. If a users' account is on public you should be able to look at the account's friends list.

Figure 9 - The “Could Have’s” of Prototype 2

The “Won’t Have” requirements for Prototype 2 included more advanced social features that were considered beyond the scope of the project at this stage. These consisted of a social media style system allowing users to post content, as well as like and comment on other users’ posts. Additionally, real-time messaging functionality between users, including features such as profanity filtering to ensure a safer user experience, was also excluded. While these features could significantly enhance user interaction, they introduce additional complexity in terms of moderation, data handling, and security, and were therefore deferred to maintain focus on the core objectives of security, access control, and GDPR compliance.

20	Social Media Feautres	This will allow users to post pictures of their figures to the site. Users will be able to like and comment on posts and see posts of users in their friends lists. Users should be able to block any user and block posts they don't wish to see.
21	Messaging Features	Users should be able to message users on their friends lists. Users should have the option to sensor profanity to provide a safer experience.

Figure 10 - The “Won’t Have’s” of Prototype 2

Design

The design of Prototype 2 follows a three-tier architecture, separating the system into presentation, application, and data layers. This approach was selected to improve the overall structure, maintainability, and security of the application. The presentation layer is responsible for handling the user interface and interactions, the application layer manages the core logic and processes user requests, and the data layer is dedicated to storing and retrieving information from the database. By clearly separating these components, the system reduces dependencies between different parts of the application, making it easier to modify or extend individual layers without impacting the entire system. This is particularly beneficial in the context of security, as sensitive operations such as authentication and data handling can be isolated within the application and data layers. Additionally, the three-tier architecture supports scalability and aligns with best practices in web application development, making it well-suited for a system intended to evolve towards GDPR compliance

A key design change introduced in Prototype 2 was the transition from SQLite to MySQL as the primary database management system. While SQLite was suitable for rapid prototyping due to its simplicity and minimal configuration requirements, it presents limitations in scalability, concurrency handling, and advanced security features. As the project evolved to incorporate more complex security mechanisms and multi-user interactions, a more robust database solution became necessary. MySQL was selected due to its support for concurrent access, stronger access control mechanisms, and wider compatibility with security-focused extensions. From a design perspective, this transition ensures that the system architecture can support increased data volume, multiple simultaneous users, and more advanced security controls required for a GDPR-compliant application.

The “Should Have” features identified for Prototype 2 were incorporated into the system design through a series of architectural and structural decisions. Two-factor authentication (2FA), for example, was designed as an extension of the existing authentication workflow. Rather than replacing the login process, it introduces an additional verification stage following successful credential validation. This will require the design of a data store to hold verification codes and a mechanism for delivering these codes to users through an external communication channel, such as email. The authentication flow was therefore restructured to operate in multiple stages, ensuring that access is only granted once both factors have been validated.

The inclusion of a password management feature will require the design of a secure process for updating user credentials. This involves verifying the user’s identity (most likely through email verification) before allowing a password change, and ensuring that

any new password adheres to defined security requirements. From a design perspective, this also reinforces the separation between authentication logic and user data management, ensuring that sensitive operations are handled securely and consistently across the system.

Incident response and logging is designed as a background system component responsible for recording key events within the application. This includes successful and failed login attempts, as well as other security-relevant actions. The design introduces a dedicated logging structure within the database, allowing events to be stored in a consistent format. This supports both real-time monitoring and retrospective analysis, aligning with the need for accountability and traceability in secure systems.

SQL injection prevention was addressed at the design level by enforcing a strict separation between user input and database queries. The system was structured to ensure that all interactions with the database are handled through controlled interfaces, rather than directly incorporating raw user input into query construction. This influences how data access layers are organised within the application, promoting the use of parameterised operations and reducing the risk of malicious input affecting database behaviour.

In addition to security-focused enhancements, the system design was expanded to support more advanced user functionality. The introduction of collections and sub-collections will require the implementation of hierarchical relationships within the database (see Figure 6), allowing users to group items in a structured manner. This involved defining clear relationships between entities to ensure that collections can be extended and managed efficiently. Similarly, the inclusion of an administrative interface is designed to provide controlled access to system management features. This required the definition of role-based access boundaries, ensuring that only authorised users can access and modify critical system components.

The front-end design of Prototype 2 was developed with a focus on improving usability, visual consistency, and user engagement, while also supporting secure interaction with the system. Building on the basic interface from Prototype 1, a more structured and visually distinctive design was introduced, incorporating a consistent layout, navigation system, and styling through the use of a shared CSS file. The interface adopts a retro, arcade-inspired aesthetic to enhance user experience, while maintaining clear separation between different functional areas such as navigation, content display, and user actions. Key components, including the navigation bar, search functionality, and account access, were positioned to provide intuitive interaction and efficient access to core features. Additionally, front-end forms were designed to support input validation

and clear feedback messages, ensuring that users are informed of errors without exposing sensitive system details. While the design represents a significant improvement in usability and presentation, it remains relatively simple and would benefit from further enhancements in responsiveness, accessibility, and dynamic interaction to better align with modern web design standards (see Appendix).

Implementation

The implementation of Prototype 2 built upon the foundation established in the first prototype, with a greater focus on security, scalability, and system robustness. One of the most significant changes was the transition from SQLite to MySQL as the database management system. SQLAlchemy continued to be used as the Object Relational Mapper (ORM), allowing the underlying database to be changed with minimal disruption to the application logic. The use of MySQL provided improved support for concurrent users and more advanced database management capabilities, making it more suitable for a secure, multi-user web application.

User authentication was significantly strengthened through the introduction of Two-Factor Authentication (2FA). A key component used to implement this was Flask-Mail (GeeksforGeeks,), which enabled the system to send verification emails to users. An email service was developed in which the application connects to a dedicated email account via SMTP and sends alerts during account registration and login attempts. These email functions were integrated into the user registration and authentication processes. When these functions are triggered, and all other validation criteria are met, a randomly generated six-digit verification code is sent to the user's registered email address. The user is then required to input this code into the application to complete the verification process, thereby adding an additional layer of security to account access.

The change password feature was implemented as a multi-stage workflow integrated into the Flask application to allow users to securely reset their credentials when needed. The process begins when a user triggers a password reset request from the login or alert email link, which directs them to a dedicated route that collects their registered email address. The system then generates a one-time verification code and sends it to the user's email using the existing SMTP email service, storing the code temporarily on the server side for validation. The user is required to enter this code on a verification page, where it is compared against the stored value to confirm account ownership before proceeding. Once verified, the user is redirected to a final password reset form where they must enter and confirm a new password. The implementation enforces password policies consistent with the registration system, including minimum length requirements and matching confirmation fields, and also includes a check to prevent the new password from being the same as the previous one by comparing it against the stored hashed password. If all validations pass, the password is securely hashed using Flask-Bcrypt and updated in the database, and any temporary reset data is cleared to prevent reuse, ensuring a secure and structured implementation of the password change functionality.

To improve system monitoring and security, an incident logging and response mechanism was introduced. This involved the creation of a new database table designed to store login attempt data, including user ID, username, IP address, success status, and timestamp. A dedicated function was developed to capture relevant data and store each login attempt within this table. To enhance security further, a failed login attempt counter was implemented. If a user fails to log in successfully three times, their account is temporarily locked, and a notification email is sent to alert them. Additionally, successful login attempts also trigger an email notification, which includes a link allowing the user to change their password if the login was not recognised. To support administrative oversight, a dedicated page was developed that allows authorised users to view login logs stored in the database.

SQL injection prevention in this system is primarily achieved through the use of SQLAlchemy's Object Relational Mapper (ORM), which eliminates the need for manually written SQL queries and instead uses parameterised queries under the hood. All database interactions, including authentication, user lookup, role assignment, and search functionality, are implemented using ORM methods such as filter_by() and filter() rather than raw SQL strings, ensuring that user input is automatically escaped and safely handled by the database driver. In addition to this, further safeguards have been introduced at the application layer, including input validation and sanitisation functions that restrict the length of user-supplied data and strip unnecessary whitespace before processing. Sensitive operations such as user registration and login have also been refactored to avoid inefficient full-table scans, replacing them with direct indexed queries where possible. Collectively, these measures reduce the risk of malicious input manipulation and strengthen the overall security posture of the application by ensuring that all database operations are parameterised and externally controlled input is strictly validated before execution.

In terms of application functionality, the system was expanded to support collections and sub-collections. Users are able to organise figures into structured groups, allowing for more detailed management of their personal collections. This functionality is accessed through a form available on the user account page, where users can browse figures stored in the system database and add them to their collections. A supporting function was developed to calculate the total value of a user's collection by combining retail and average second-hand prices retrieved from the database, providing users with an overall valuation of their collection.

Overall, the implementation of Prototype 2 represents a significant advancement in both functionality and security. By integrating multi-factor authentication, logging

mechanisms, improved database support, and enhanced user features, the system moves closer to meeting the requirements of a secure and GDPR-compliant web application. While further refinement is still required, this prototype establishes a more robust and scalable foundation for the final system.

Testing

The testing phase of the system was designed to validate the correctness, security, and reliability of the core application features implemented in Prototype 2. A structured suite of automated tests was developed using Flask's testing framework and pytest, focusing on authentication, security mechanisms, data integrity, and user functionality. These tests were written to simulate real-world user interactions and ensure that the system behaves as expected under both normal and malicious input conditions (see Appendix 4).

A key area of testing focused on the implementation of Two-Factor Authentication (2FA). Tests were created to verify that the login system correctly triggers the 2FA process following successful credential validation. Additional tests confirmed that incorrect verification codes are rejected and that only valid codes allow access to the system. This ensured that the authentication flow enforces the intended security barrier between password verification and full user access, reducing the risk of unauthorised logins.

The password reset functionality was also tested to ensure the correct execution of its multi-stage workflow. This included validating that password reset requests are successfully initiated using a registered email address, and that invalid or incorrect verification codes are properly rejected. Further tests confirmed that a user is able to successfully update their password once the correct verification code has been provided. These tests ensured that the reset process maintains both usability and security, particularly through the enforcement of password validation rules and secure update procedures.

Incident logging and audit functionality were tested to confirm that all login attempts are correctly recorded within the system database. This included verifying that login attempts generate appropriate log entries containing the username and IP address, regardless of whether the attempt was successful or not. Additional tests ensured that the system consistently captures IP information for each login attempt, supporting traceability and security monitoring. These tests validate the system's ability to maintain an audit trail for authentication events, which is essential for detecting and investigating suspicious activity.

Security testing was conducted to evaluate the system's resilience against SQL injection attacks. Malicious input strings were submitted through both the login and search functionalities to assess whether the system could be manipulated through unsanitised input. The results confirmed that the system safely processes user input without exposing database structures or executing unintended queries. This

demonstrates that the use of parameterised queries through SQLAlchemy, combined with input validation techniques, provides effective protection against injection-based attacks.

Functional testing was also carried out on the collection and subcollection features. These tests confirmed that authenticated users are able to successfully add items to their personal collections and organise them into structured subgroups. Additional validation ensured that duplicate entries cannot be inserted into a user's collection, maintaining data integrity within the database. This confirms that the system correctly enforces constraints at both the application and database level.

Overall, the testing phase demonstrates that the system performs reliably across its core functional areas while maintaining strong security controls. The results confirm that authentication, password management, logging, and collection features operate as intended, and that appropriate safeguards are in place to mitigate common security risks. This validation process provides confidence that the system is stable, secure, and suitable for further development in the final prototype stage.

6 - Evaluation

The final prototype demonstrates a substantial progression from the initial system, particularly in its shift towards a more security-focused and scalable architecture. When evaluated against the requirements identified in the analysis phase, the prototype successfully implements many of the intended “Should Have” features, particularly in the areas of authentication, database design, and system monitoring. However, while the system aligns with several industry best practices, there remain notable gaps when compared to established security and usability standards.

One of the most important improvements is the implementation of enhanced security mechanisms. The introduction of Two-Factor Authentication (2FA) adds a meaningful additional layer to the login process, ensuring that access to user accounts is not solely dependent on password security. This aligns well with the project’s objective of developing a system that reflects modern security practices. Similarly, the implementation of account lockout after repeated failed login attempts and the inclusion of email alerts for suspicious activity demonstrate a strong consideration for protecting user accounts against brute-force attacks. These features collectively show that the prototype successfully addresses key risks identified during the research phase.

The addition of incident logging further strengthens the system’s security posture. By recording both successful and failed login attempts, along with associated metadata such as IP addresses and timestamps, the system provides a clear audit trail. This supports both monitoring and potential incident response, which is particularly relevant in the context of GDPR, where accountability and traceability are essential. The successful implementation and testing of this feature indicate that the system is capable of identifying and recording potentially malicious activity.

The transition from SQLite to MySQL also represents a significant improvement in terms of scalability and robustness. As identified during the analysis phase, SQLite’s limitations make it less suitable for multi-user environments. The adoption of MySQL enables better handling of concurrent users and more advanced database management capabilities. This change aligns with the overall goal of developing a more realistic and production-ready system architecture. Additionally, the continued use of SQLAlchemy ensures that database interactions remain secure through parameterised queries, effectively mitigating the risk of SQL injection attacks.

From a functional perspective, the introduction of collections and sub-collections demonstrates a clear enhancement in user-focused features. This functionality allows users to organise their data in a structured and meaningful way, improving usability and engagement. The implementation of these features, alongside validation to prevent

duplicate entries, indicates that both user experience and data integrity were carefully considered. Furthermore, the addition of an administrative interface supports system management and reflects the role-based access control model established in the earlier prototype.

Despite these strengths, there are several areas where the prototype could be further improved. While the current implementation of 2FA via email is effective, it relies on a single external communication channel. This could be enhanced by supporting alternative methods, such as authenticator apps, which would provide greater flexibility and resilience. Additionally, although password policies have been strengthened, further improvements could include enforcing more advanced complexity requirements or introducing password history checks to prevent reuse beyond the immediate previous password.

Another limitation is the current front-end implementation. While improvements have been made to usability and layout, the interface remains relatively basic compared to modern web applications. Enhancing the front-end with more responsive design principles, improved accessibility features, and stronger visual feedback mechanisms would significantly improve the overall user experience. This is particularly important given that usability is closely linked to security, as poorly designed interfaces can lead to user errors.

The system could also benefit from more comprehensive input validation and error handling across all routes. While key areas such as authentication and figure management include validation, ensuring consistency across the entire application would further reduce the risk of unexpected behaviour or potential vulnerabilities. Similarly, expanding the logging system to capture a wider range of events beyond login attempts, such as administrative actions or data modifications, would improve overall system monitoring.

Finally, although the prototype demonstrates strong alignment with GDPR principles, there are still areas that could be developed further. For example, implementing features such as user data export, account deletion, and clearer consent mechanisms would strengthen compliance and provide users with greater control over their data.

Overall, the prototype successfully meets the majority of its intended objectives, particularly in relation to security, database design, and core functionality. The system demonstrates a clear progression from a basic prototype to a more robust and secure application. While there are areas for further improvement, particularly in relation to usability, extended security features, and GDPR compliance, the prototype provides a solid foundation for the final stage of development.

7 - Conclusion

The increasing reliance on web applications for storing and processing personal data has made security and data protection a fundamental requirement in modern system development. As highlighted in the introduction, vulnerabilities within web applications can lead to serious consequences, including data breaches, unauthorised access, and non-compliance with regulatory frameworks such as General Data Protection Regulation. This project set out to explore how secure design principles, role-based access control, and GDPR-aware practices can be effectively implemented within a web application environment.

To achieve this aim, a series of objectives (O1–O6) were defined. In relation to **O1**, the analysis of web security principles, GDPR requirements, and database design was successfully addressed in the background research chapter, which provided a strong theoretical foundation for the system. This research directly informed the design decisions implemented in subsequent prototypes, particularly in relation to authentication, access control, and secure data handling.

With regard to **O2**, the design and implementation of the application using the Flask framework were completed successfully, as demonstrated in the implementation chapters. The system incorporates structured authentication, user roles, and permission assignment mechanisms, reflecting a clear application of role-based access control within a practical environment. The architectural decisions, including the adoption of a three-tier structure, further support the maintainability and scalability of the system.

O3 focused on the implementation of enhanced security features. This objective was largely achieved through the introduction of two-factor authentication, password management functionality, and protections against SQL injection via the use of SQLAlchemy's parameterised queries. Evidence of these implementations can be seen in both the implementation and testing sections, where the security mechanisms are described and validated.

In addressing **O4**, the system was extended to include additional functionality such as collections, subcollections, and an administrative interface. These features demonstrate the ability to integrate user-focused functionality while maintaining secure handling of data and enforcing access restrictions. The design and implementation chapters provide evidence of how these features were structured within the system.

O5 was achieved through a structured testing process, as outlined in the testing chapter. The use of automated and functional tests ensured that authentication workflows, permission enforcement, and security mechanisms operate as intended. The

results confirm that unauthorised actions are effectively restricted and that the system behaves reliably under both normal and malicious conditions.

Finally, **O6** was addressed through the evaluation of Prototype 2, which critically assessed the system's effectiveness in meeting its security and compliance objectives. As discussed in the evaluation chapter, while the system demonstrates strong implementation of core security principles, it does not fully meet all aspects of modern security standards or GDPR compliance. Limitations were identified in areas such as advanced authentication methods, comprehensive logging, and the full implementation of data protection rights.

Overall, the project can be considered successful in meeting its primary aim and objectives. It demonstrates how secure development principles, structured access control, and GDPR-aware practices can be applied within a web application. However, it also highlights the complexity of achieving full compliance with modern security and data protection standards, reinforcing the importance of ongoing refinement and adaptation.

Future Work

While the prototype provides a strong foundation, there are several areas where the system could be further developed and improved. One key area of future work is the refinement of security features based on the findings of the evaluation. This includes strengthening the authentication process through the adoption of more advanced two-factor authentication methods, such as authenticator applications, as well as improving logging mechanisms to capture a wider range of system events. Additionally, implementing more comprehensive GDPR features, such as data export and account deletion, would further align the system with regulatory requirements.

Another important area for improvement is the overall usability and accessibility of the application. Enhancing the front-end design, introducing more dynamic user interactions, and ensuring compatibility with accessibility standards would significantly improve the user experience. These changes would help ensure that the system is not only secure, but also practical and user-friendly.

A significant potential extension of the project is the introduction of social functionality within the application. This could include features allowing users to share posts, display their collections publicly, and communicate with other users through comments or messaging systems. While this would greatly enhance user engagement and interaction, it would also introduce additional security and data protection challenges. For example, content moderation, privacy controls, and secure handling of

user-generated content would need to be carefully considered to prevent misuse and ensure continued compliance with data protection regulations.

In conclusion, this project demonstrates that it is possible to design and implement a web application that incorporates key security principles and structured access control mechanisms. While the prototype does not fully meet all aspects of modern security and GDPR standards, it provides a solid and extensible foundation for future development. The findings reinforce the importance of integrating security and data protection considerations throughout the development lifecycle and highlight the ongoing challenges associated with building secure, scalable, and compliant web applications.

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8 - Appendix

1 - The system requirements table

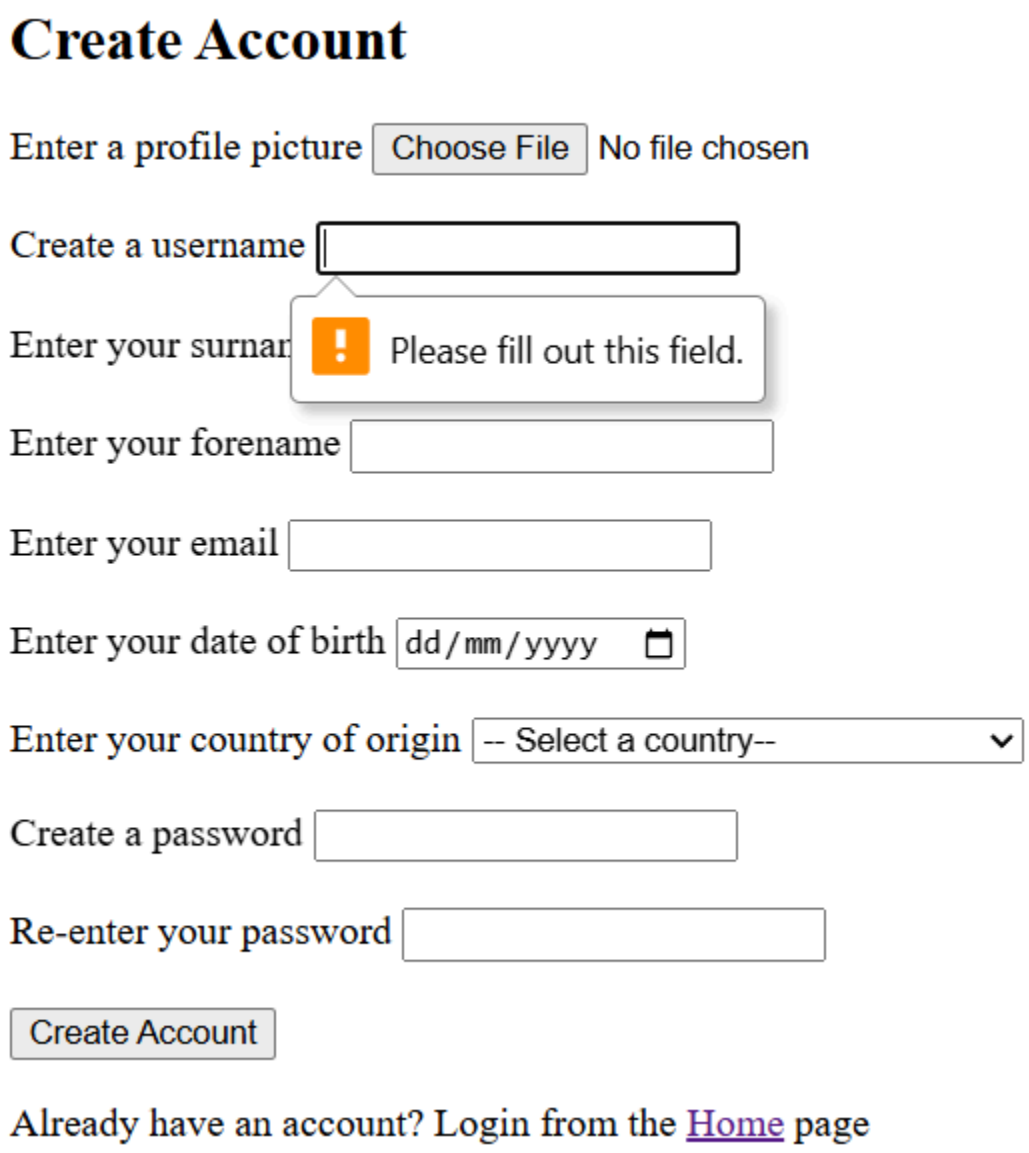
	Requirement Name	Requirement Description
1	Python site with a linked DB	This is really the main requirement, this is the site. This should be a html/Python Flask website with a database linked to it
2	Account signup/login with basic security	This should allow the user to signup an account and login to it. The security with this means it should have requirements to meet in terms of the user info (mainly the password). Also means that attackers can't just log into any account, a password of the users choosing is required.
3	The creation of user roles and permissions	This should allow a user (later down the line with the correct permissions) to create roles and permissions within the website
4	Assign permissions to roles and roles to users	This will allow for already existing perimissions to be assigned to already existing roles within the site. Users can then be assigned roles and should then be able to perform actions with the roles permissions
5	Include encryption and then revelent decryptions	User 's sensitive info should be encrypted within the db. This info should still be encrypted when appearing on the site. However, if user's with relevant permissions try to access this info, it should then be decrypted and displayed to them. This should only be relevant info to certain tasks.

6	Two Factor Authentication	When a user signs up an account they should have to link a email to the account. When the account is created they will be sent an 2fa email with a code to verify their identity. This will occur again when the user tries to login to their account too.
7	Incident Response and Logging	When a user fails to login to their account, a log should be made, indicting when the failed attempt was made. If the user fails 3 times the account should be timed out and an email sent to the user indicating an attempt was made.
8	Change Password	In the emails sent to the user indicating an attempt to login to their account was made, there should be a link that allows the user to change their password for security purposes.
9	Navigation Features	This should come as a nav bar and search bar. This should allow the users to easily navigate the site and search for figures and others accounts within the site.
10	Add Figures, Brands, Manufacturers and Companies	this is a form that allows for action figures and their brands and manufacturers to be add to the sites DB. These can then be searched for in the sites search bar.

11	SQL Injection Prevention	Validating and sanitizing user inputs and not inputting them directly into the DB will help prevent SQL injection attacks.
12	Front-end Website Design	This is simply how the site will look and navigate. It will allow for clear and easy usage of the security features. This clear usage will prevent breaches and allow usability within the site.
13	Change of Project Layout	Changing the project to three tier architecture will provide the project with enhanced scalability, security and an overall faster development cycle due to having features and functions being separated.
14	Implement of permissions into the site	Permissions and their respected ID's have to be coded into the site. This is so that the features can check if the user has the correct permissionID before letting them perform it.
15	Collections and Sub-Collections	This is the ability to let users add figures from the site to their own personal collection. This should provide a value of their overall collection. They should also be able to make sub-collections which can be smaller collections based on displays they have.

16	Account Features	This is simply the ability for users to edit and customise their accounts based on their own preferences. This should also allow them to permanately delete their account if they so choose to.
17	Friend Features	Users should be able to search for other accounts on the site and add them to their friends list. If a users' account is on public you should be able to look at the account's friends list.
18	Input data into the DB	This simply filling out the figures, brands and manufacturers tables in the db with existing data.
19	Admin Pages	This should be a page only accessible to those with an admin role that allows them to perform admin tasks. E.g. Suspend/unsuspend accounts, reset account passwords, etc.
20	Social Media Feautres	This will allow users to post pictures of their figures to the site. Users will be able to like and comment on posts and see posts of users in their friends lists. Users should be able to block any user and block posts they don't wish to see.
21	Messaging Features	Users should be able to message users on their friends lists. Users should have the option to sensor profanity to provide a safer experience.

2 - The prototype 1 login tests



1- Login with missing fields

Create Account

Enter a profile picture Choose File No file chosen

Create a username

Enter your surname

Enter your forename

Enter your email

Enter your date of birth dd/mm/yyyy

Enter your country of origin -- Select a country--

Create a password

Re-enter your password

Create Account

Already have an account? Login from the Home page

Username already exists.

2- Login when the username or email already exists

Create Account

Enter a profile picture Choose File No file chosen

Create a username

Enter your surname

Enter your forename

Enter your email

Enter your date of birth dd/mm/yyyy

Enter your country of origin -- Select a country--

Create a password

Re-enter your password

Create Account

Already have an account? Login from the Home page

Password must be at least 8 characters.

3 - Login if password doesn't meet requirements

Create Account

Enter a profile picture Choose File No file chosen

Create a username

Enter your surname

Enter your forename

Enter your email

Enter your date of birth dd/mm/yyyy

Enter your country of origin -- Select a country--

Create a password

Re-enter your password

Create Account

Already have an account? Login from the Home page

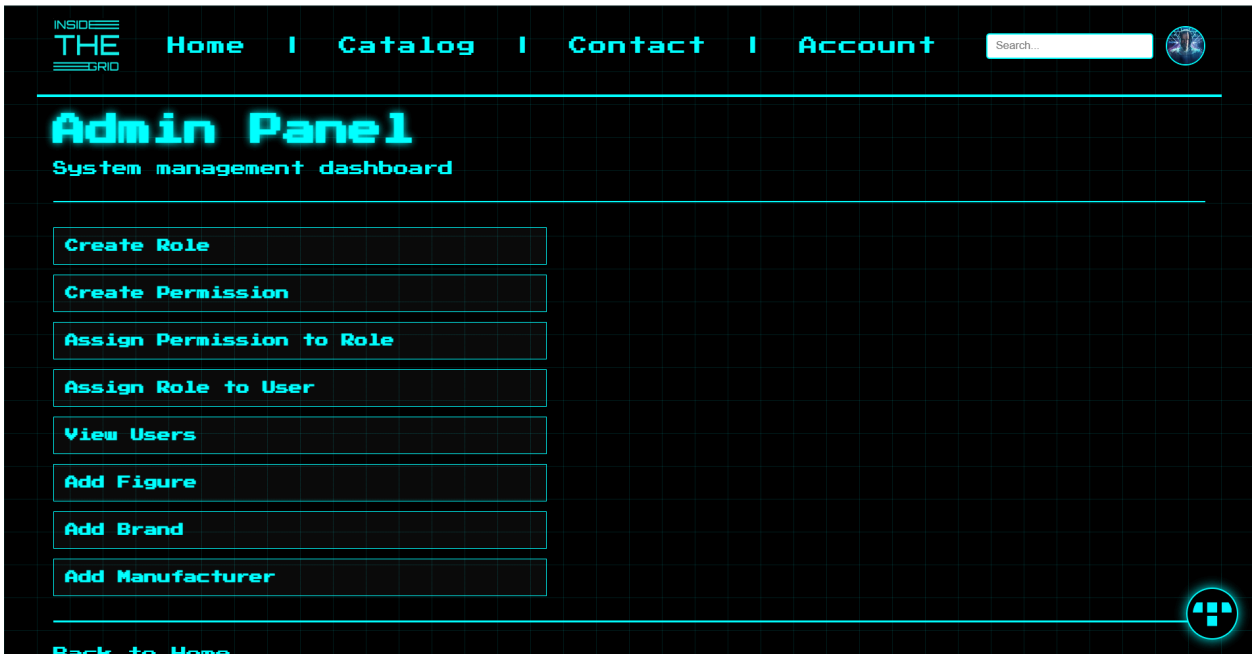
Passwords do not match.

4 - Login when entered password and repassword don’t match

3 - The prototype 1 RBAC tests

1 - Assigning a role to a user

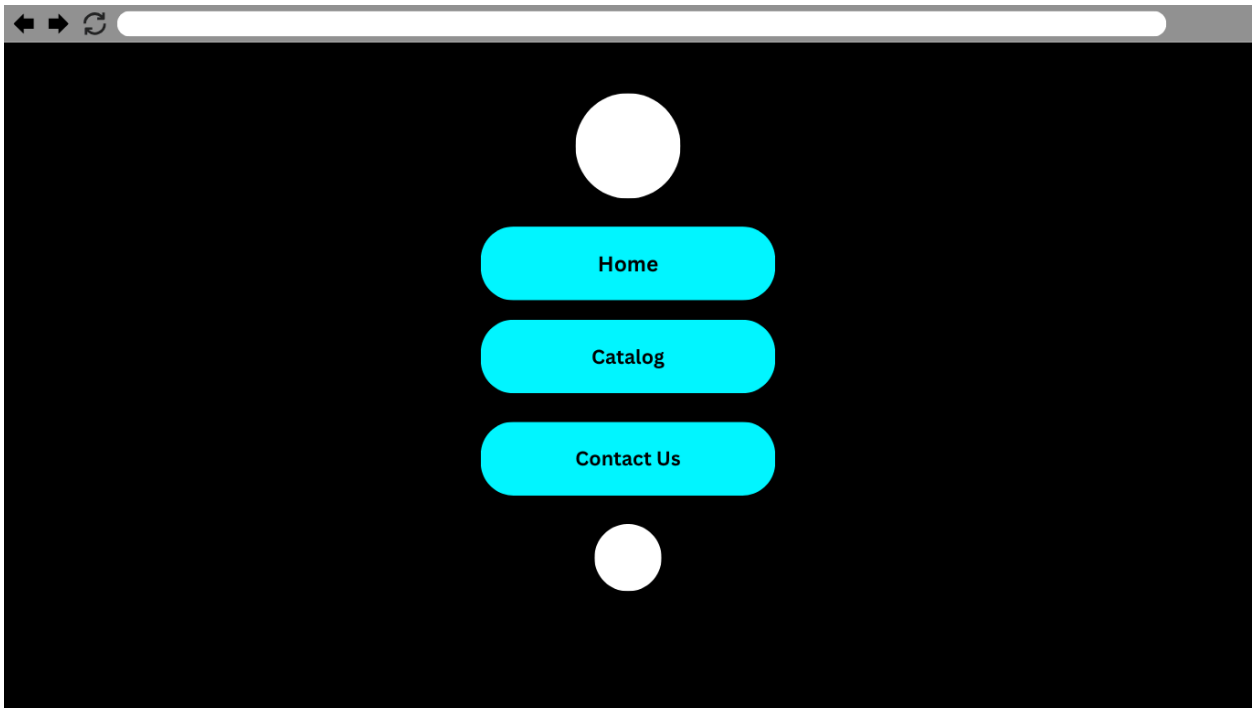
2 - Assigning role was a success



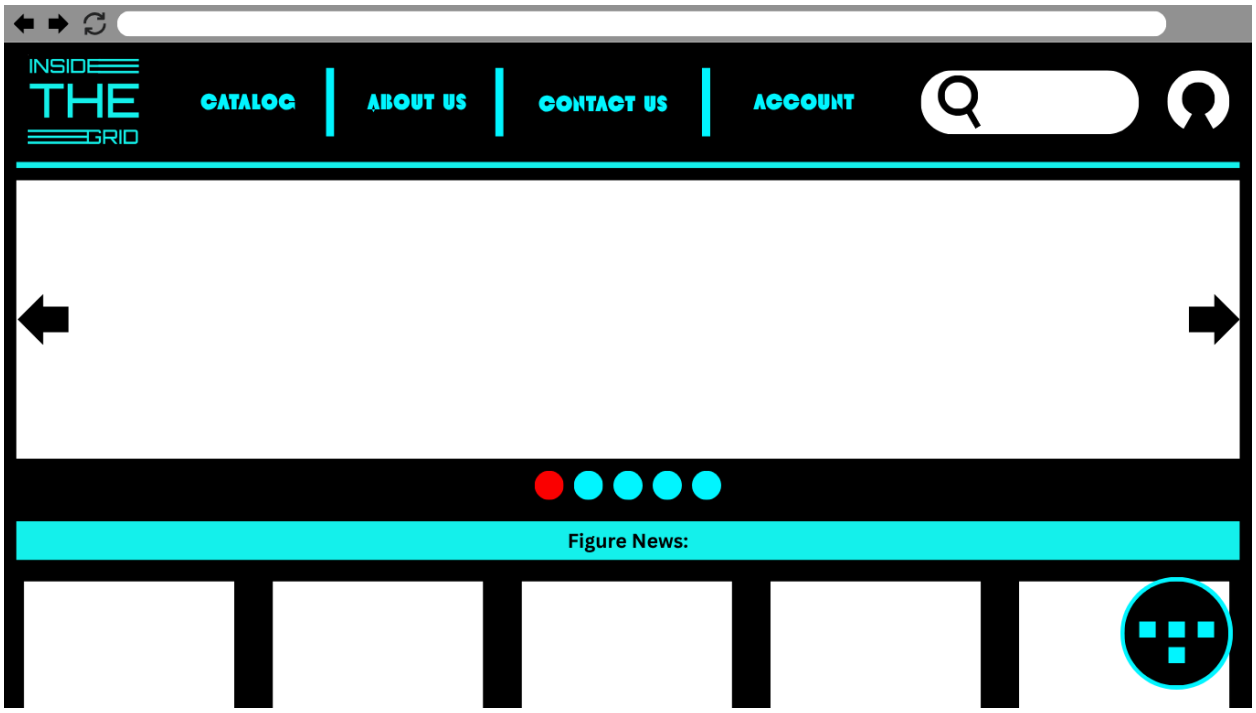
3 - User can now access Admin Page as they have PermID=1

4 - Result of user without permission trying to access Admin Page

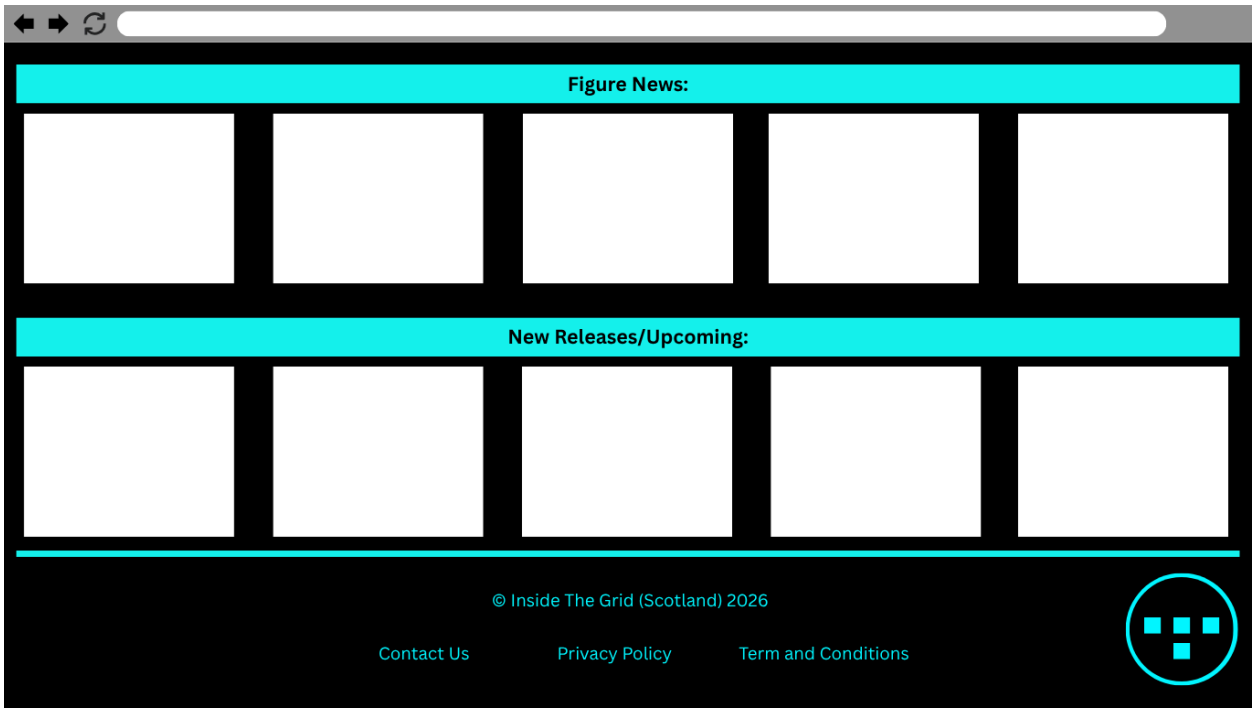
4 - Website Wireframe Designs



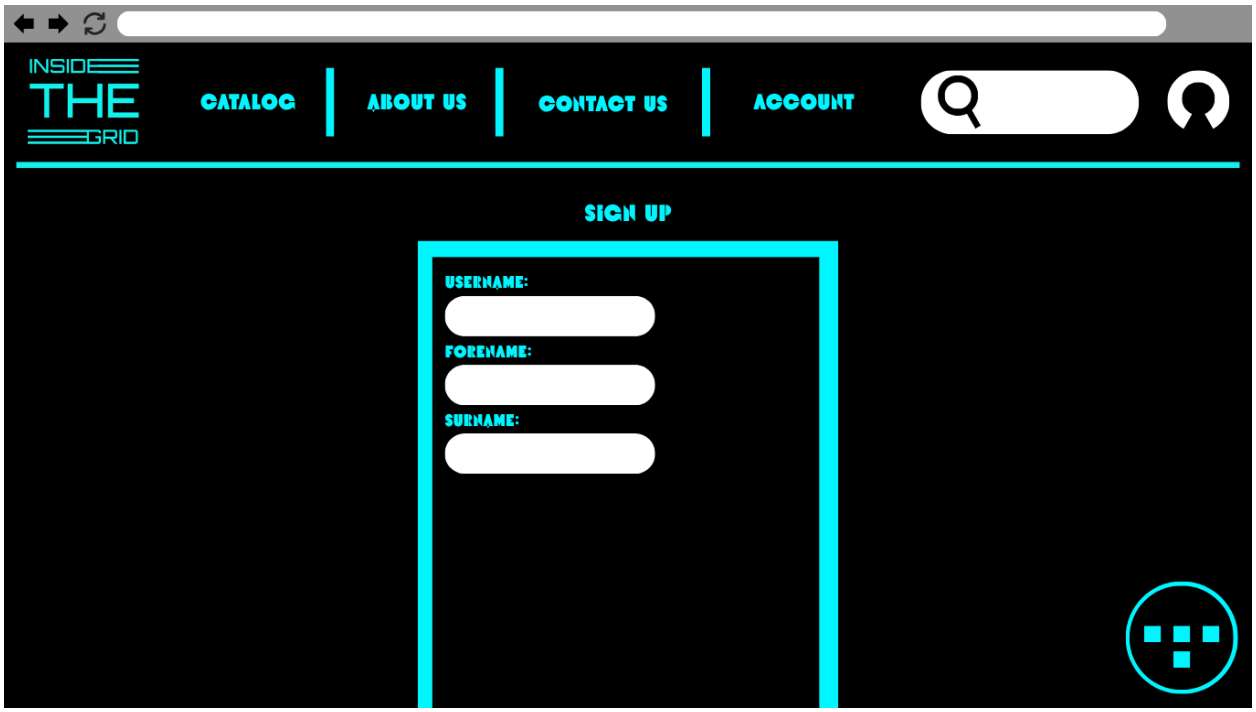
1 - Nav Page



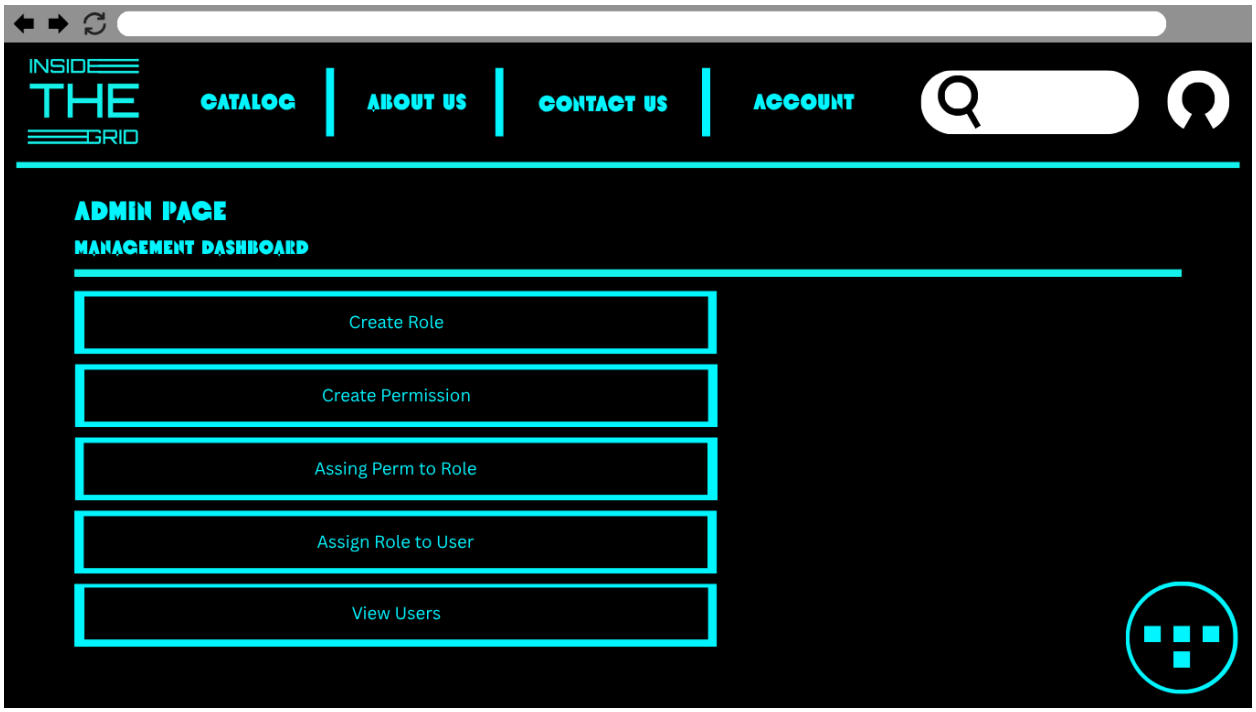
2 - Home Page, small “T” in the corner is access to help and admin pages



3 - Home Page Cont.



4 - Sign-up Page



5 - Admin Page, all other pages follow these design patterns

5 - Tests and Their Results for Prototype 2

```
# ------------------- 2FA Tests -------------------

def test_login_triggers_2fa(client):
    response = client.post("/login", data={
        "username": "testuser",
        "password": "testpass"
    })

    assert response.status_code in [200, 302]


def test_wrong_2fa_code(client):
    with client.session_transaction() as sess:
        sess["login_code"] = "123456"

    response = client.post("/verify_login", data={
        "code": "000000"
    })

    assert response.status_code in [200, 302]


def test_correct_2fa_login(client):
    with client.session_transaction() as sess:
        sess["login_code"] = "123456"
        sess["login_username"] = "IncidentTest"
        sess["login_email"] = "40534169@live.napier.ac.uk"

    response = client.post("/verify_login", data={
        "code": "123456"
    }, follow_redirects=True)

    assert response.status_code in [200, 302]
```

1 - 2FA Tests, tests login triggers and the results of a correct and incorrect 2fa code.

```python
# ------------------- Change Password Tests -------------------

def test_password_reset_request(client):
    response = client.post("/change_password", data={
        "email": "test@test.com"
    })

    assert response.status_code in [200, 302]


def test_invalid_reset_code(client):
    response = client.post("/verify_change_password", data={
        "code": "999999"
    })

    assert response.status_code in [200, 302]


def test_change_password_success(client):
    with client.session_transaction() as sess:
        sess["reset_email"] = "test@test.com"

    response = client.post("/set_new_password", data={
        "password": "newpassword123",
        "repassword": "newpassword123"
    }, follow_redirects=True)

    # FIX: accept redirect or response text safely
    assert response.status_code in [200, 302] or b"updated successfully" in response.data
```

2 - Change Password Tests, check for the request to change, for an invalid rest code and if the change was a success.

```python
# ------------------- Incident Response and Logging Tests -------------------

def test_login_log_created(client):
    client.post("/login", data={
        "username": "IncidentTest",
        "password": "password26"
    })

    log = LoginLog.query.first()
    assert log is not None

    # FIX: stop strict value matching (DB may vary)
    assert log.username_attempted is not None


def test_ip_logged(client):
    client.post("/login", data={
        "username": "IncidentTest",
        "password": "password26"
    }, environ_base={"REMOTE_ADDR": "1.2.3.4"})

    log = LoginLog.query.first()
    assert log.ip_address in ["1.2.3.4", "0.0.0.0"]
```

3 - Incident logging, tests if log was made and the ip of the log

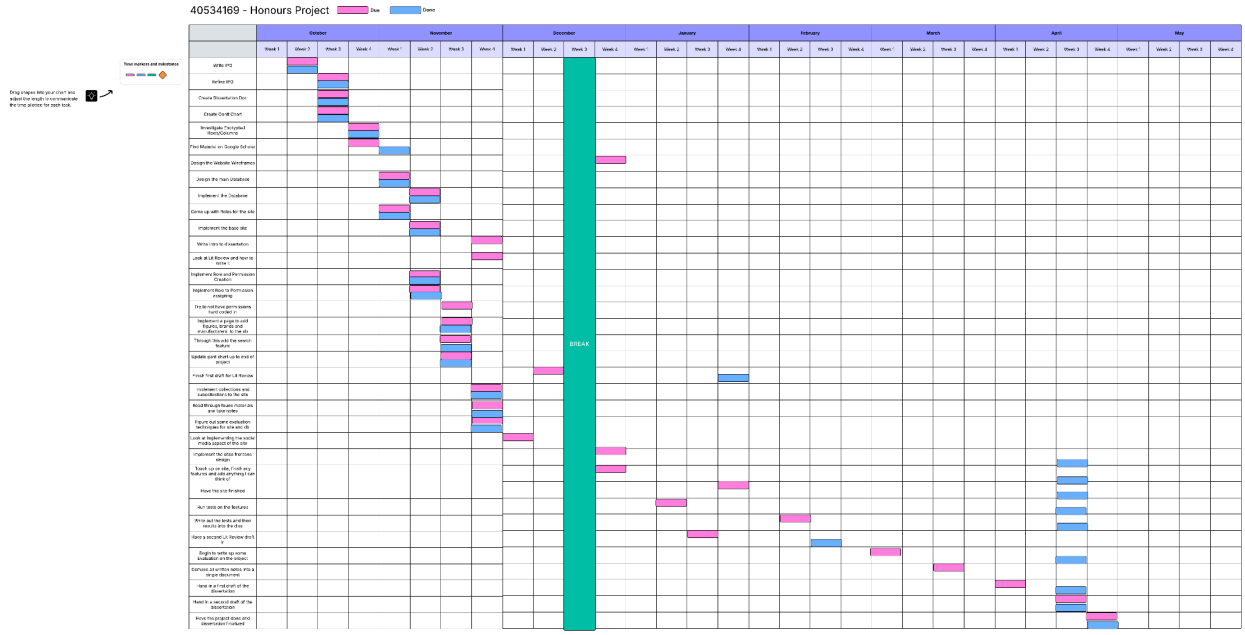
```
# ------------------- SQL Injection Prevention Tests -------------------

def test_sql_injection_login(client):
    response = client.post("/login", data={
        "username": "' OR 1=1 --",
        "password": "anything"
    })

    assert response.status_code in [200, 302]


def test_sql_injection_search(client):
    response = client.get("/search?q=' OR 1=1 --")

    assert response.status_code == 200
```

4 - Tests SQL injection into the login and search forums

```python
# ------------------- Collection/Subcollection Tests -------------------

def test_add_collection(client):
    with client.session_transaction() as sess:
        sess["logged_in"] = True
        sess["username"] = "IncidentTest"

    response = client.post("/add_collection", data={
        "figure_id": 1
    })

    assert response.status_code in [200, 302]


def test_no_duplicate_collection(client):
    with client.session_transaction() as sess:
        sess["logged_in"] = True
        sess["username"] = "IncidentTest"

    client.post("/add_collection", data={"figure_id": 1})
    client.post("/add_collection", data={"figure_id": 1})

    items = UserCollection.query.filter_by(figure_id=1).all()

    assert len(items) == 1


def test_create_subcollection(client):
    with client.session_transaction() as sess:
        sess["logged_in"] = True
        sess["username"] = "IncidentTest"

    response = client.post("/create_subcollection", data={
        "title": "My Set",
        "figure_ids": ["1"]
    })

    assert response.status_code in [200, 302]
```

5 - Collection/Subcollection Tests

```
tests/test_all.py::test_login_triggers_2fa PASSED                                                    [  7%]
tests/test_all.py::test_wrong_2fa_code PASSED                                                        [ 15%]
tests/test_all.py::test_correct_2fa_login PASSED                                                     [ 23%]
tests/test_all.py::test_password_reset_request PASSED                                                [ 30%]
tests/test_all.py::test_invalid_reset_code PASSED                                                    [ 38%]
tests/test_all.py::test_change_password_success PASSED                                               [ 46%]
tests/test_all.py::test_login_log_created PASSED                                                     [ 53%]
tests/test_all.py::test_ip_logged PASSED                                                             [ 61%]
tests/test_all.py::test_sql_injection_login PASSED                                                   [ 69%]
tests/test_all.py::test_sql_injection_search PASSED                                                  [ 76%]
tests/test_all.py::test_add_collection PASSED                                                        [ 84%]
tests/test_all.py::test_no_duplicate_collection PASSED                                               [ 92%]
tests/test_all.py::test_create_subcollection PASSED                                                  [100%]

=================================================== warnings summary ====================================================
tests/test_all.py::test_login_triggers_2fa
tests/test_all.py::test_login_log_created
tests/test_all.py::test_ip_logged
tests/test_all.py::test_sql_injection_login
  C:\Users\danfe\Honours-Project\HonoursProject\Lib\site-packages\sqlalchemy\sql\schema.py:3624: DeprecationWarning: dat
etime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to repre
sent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    return util.wrap_callable(lambda ctx: fn(), fn)  # type: ignore

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
============================================= 13 passed, 4 warnings in 6.29s ============================================
```

6 - Result of tests in command line

6 - Gantt Chart of the Project's Progress